



OPEN Rice straw and single-use plastic waste co-pyrolyzed plastic-char composite for the partial replacement of cement in concrete: a life cycle perspective

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This study examines the valorization of single-use plastic and rice straw waste via co-pyrolysis to produce a plastic-char composite (PRSB) and rice straw biochar (RSB) for sustainable building applications. The generated biochars were thoroughly analyzed for their physicochemical characteristics using SEM, FTIR, XRD, and Raman spectroscopy. RSB and PRSB were utilized as partial cement substitutes (1–7% w/w) in M30-grade concrete to produce high-strength and durable composites. Optimization of mechanical performance was executed via a Box–Behnken design, focusing on compressive, split tensile, and flexural strength, with curing time as a critical variable. Experimental results indicated that a substitution of up to 3% markedly improved mechanical characteristics relative to ordinary concrete. The algorithm forecasted an ideal combination with 3.8% substitution and 28 days of curing, optimizing strength across all metrics. Sustainability was evaluated by a cradle-to-gate Life Cycle Assessment (LCA). With a 1% substitution of PRSB, the Global Warming capacity (GWP 500) from fossil emissions decreased from 80% to 72%, while biogenic emissions declined from 78% to 60%. However, substitution levels over 5% elevated total GWP due to increased processing, resource, and durability-related stresses. The findings indicate that minimal PRSB integration (< 5%) can enhance mechanical performance while providing substantial environmental benefits, thereby promoting the circular economy by upcycling agricultural and plastic waste into sustainable, low-carbon concrete.

Keywords Biochar, Plastic-char composite, Mechanical strength, Cement and rice straw

The rapid increase in demand for new construction has rendered the cement manufacturing process a significant issue due to its substantial contribution to atmospheric CO₂ emissions¹. Cement is an important construction material required for construction work, despite having numerous environmental issues. Cement is the predominant cementitious material utilized in infrastructure development and building construction, owing to its strength, cost-effectiveness, and compatibility with other construction materials. India ranks second globally as a cement manufacturer with an approximate 3.96 billion tonnes of cement production in the year 2023–2024². Cement manufacturing is an energy-intensive process that uses non-renewable energy resources, accounting for up to 9% of global greenhouse gas emissions³, which are anticipated to rise to 23% by 2050⁴. Furthermore, single-use plastic, commonly known as low-density polyethylene (LDPE), is a huge issue that the globe is dealing with because it is non-recyclable⁵. Approximately 359 million metric tonnes of single-use plastic waste was generated globally in 2021⁶, accounting for 3.4% of global greenhouse gas emissions⁷. In addition, India is dealing with an air pollution problem caused by open stubble burning in the northern part of the country⁸.

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Concrete is commonly used in construction because of its strength, durability, and adaptability. However, over time, concrete can develop a variety of technical issues, including cracking, corrosion, alkali-aggregate reaction (which causes swelling), freeze-thaw damage in cold climates, sulfate attack, abrasion, and wear. Various waste materials, such as fly ash, slag waste, and ground granulated blast furnace slag (GGBS), were used as substitutes for cement in the concrete mix to overcome the mechanical limitations of conventional concrete^{9–12}. However, there are specific problems associated with these replacement materials such as, variable strength at higher replacement percentages, high settling time, increased water demand, and poor durability in cold climatic conditions^{13–16}. Likewise, utilization of blast furnace waste is often linked to variability in chemical properties and particle size distribution, reduced early strength, slower hydration, and leachability of harmful toxins and heavy metals into the environment^{12,17}.

In this regard, biomass-derived porous carbon material, called biochar, has gained significant attention in the last few decades for its application in developing sustainable concrete¹⁸. For instance, Maljaee et al.¹⁹ examined how partially replacing cement with biochars of different chemical compositions affected cement mortar performance. Biochars replaced 0.5, 1, 2, and 4 wt% of cement. The study found that biochars improved hydration products. The 28-day compressive strength increased somewhat using 4 wt% olive stone and rice husk biochars as cement substitutes. A pozzolanic reaction in the matrix gave rice husk biochar improved long-term mechanical performance¹⁹. Another study by Chen et al.²⁰ used sludge-derived biochar as an additive material to replace cement @ 2 wt% in the concrete mix. The results indicated that the incorporation of sludge-derived biochar in cement composite slightly reduced its mobility; however, it increased its 1-day and 28-day strength by 29% and 5.8%, respectively. Mosaberpanah et al.²¹ conducted a study on the effect of the co-application of biochar and sewage sludge ash as a partial replacement material on the mechanical properties and durability of cementitious composites. The biochar and sewage sludge ash were applied at a rate of 6% and 10%, respectively, and the mechanical strength and durability parameters were studied for the curing age of 7 and 28 days²¹. The results showed that the average compressive strength of the cubes increased by 21.68% at the end of 28 days. In another investigation, Gupta et al.²² examined biochar and silica fumes as partial replacements for cement in mortar. Biochar and silica fumes increased compressive strength by 18–20% over normal cement mortar. The water permeability test showed that biochar and silica fumes decreased capillary water absorption by 50–60%, enhancing cement mortar durability²². Another widely cited study comes from Gupta and Kashani²³. This study produced biochar from peanut shells and demonstrated that adding 1 wt% and 3 wt% biochar enhanced the 1-day, 3-day, and 7-day compressive strengths by 18–20% compared to plain mortar. Critically, when 3% biochar was combined with 20% fly ash, the 7-day strength improved by 19% compared to using only 20% fly ash alone, demonstrating how biochar augmentation improved the performance what fly ash achieves in the same system²³. Gupta et al.²⁴ investigated the combination of biochar and fly ash in cement mortar under accelerated carbonation curing. The findings indicated that the addition of 3 wt% biochar from peanut shells increases the carbonation of cement mortar and also aids in the production of hydration products with a higher degree of hydration than the control. The addition of biochar to the fly ash mixture also reduces the rate of water absorption, thus improving the durability of concrete. Thus, all previous studies showed that the biochar application in the cementitious materials is a viable alternative to produce green concrete mix with improved mechanical performance.

Novelty of the study

Recently, porous carbon materials, including biochar, have been identified as promising partial substitutes for cement in concrete and mortars due to their ability to optimize pore structure and enhance durability. However, all current studies focus on biomass-based biochar, and the use of plastic waste, particularly carbonized plastic waste, as a cement substitute has yet to be investigated. This paper fills this critical gap by proposing a new thermochemical method that combines agricultural residues and single-use plastic waste via controlled co-pyrolysis, enabling the valorization of waste while reducing cement. In this study, rice straw and single-use plastic waste were co-pyrolyzed in a nitrogen-purged furnace to create a unique porous plastic-char composite. The plastic-derived carbon phase promotes structural densification, while the biomass-derived carbon promotes porosity and interfacial bonding. The plastic-char composite was used as a partial substitute for cement in M30-grade concrete and its mechanical properties were assessed. A cradle-to-gate Life Cycle Assessment (LCA) was conducted to assess the environmental impact of the proposed material system, along with its mechanical properties, and to compare it with conventional concrete. This paper applies carbon synthesis from waste, concrete use, and life cycle assessment validation to support innovation in materials, the responsible use of waste, and environmentally sustainable construction practices, in line with Sustainable Development Goals 9, 12, and 13. The target 9.4 of SDG 9 aims to improve infrastructure and transform industries to be sustainable, with higher resource efficiency and the use of clean technologies. The developed thermochemical conversion technology turns agricultural and plastic waste into useful cementitious additives, hence increasing efficiency in the construction sector. This is a direct contribution to Indicator 9.4.1 (CO₂ emissions per unit of value added), as it reduces the use of clinker, the most emission-intensive material in concrete. The research also contributes to Target 12.5 of SDG 12, which seeks to reduce waste generation through prevention, reduction, recycling, and reuse. The use of rice straw and plastic waste to produce a biochar composite material gives these materials a second chance rather than disposing of them by open burning and landfilling. This supports Indicator 12.5.1 (National recycling rate and tons of material recycled) in showing how the valorization of industrial waste can be achieved in the construction industry. The connection to climate action is found under Target 13.2. The cement industry accounts for about 7–8% of total CO₂ emissions worldwide (IPCC AR6). By partially replacing clinker and reducing the global warming potential in the LCA study, the proposed material actively helps in climate change mitigation, as it satisfies Indicator 13.2.2 (Total greenhouse gas emissions per year). The reduction in GWP, fossil fuel consumption, and particulate matter emissions helps in quantifying climate change mitigation.

Type of material	C %	N %	H %	O %	O/C	H/C	VM %	Ash %	FC %
Rice Straw	37.48	5.21	2.68	54.63	1.094	0.851	54.83	18.72	16.45
Low-density Plastic	83.76	4.36	8.14	3.74	0.033	1.157	97.11	2.33	0.56
Plastic-char composite	74.39	2.59	4.84	18.18	0.183	0.775	64.33	13.76	21.91
Biochar	64.44	2.24	2.96	30.36	0.353	0.547	47.32	32.48	18.20

Table 1. Elemental and proximate analysis of plastic-char composite.

Biochar type	BET Surface area (m ² /g)	Total pore volume (cm ³ /g)	Average pore diameter (nm)	Particle size distribution	True density (g cm ⁻³)
RSB	11.06	0.042	8.63	D ₅₀ ~ 90–120 µm D ₉₀ ~ 280–340 µm	0.72 ± 0.05
PRSB	37.31	0.027	2.306	D ₅₀ ~ 70–80 µm D ₉₀ ~ 150–180 µm	0.86 ± 0.04

Table 2. BET surface area analysis data.

Results and discussion

Chemical characteristics of plastic-char composite

Elemental and proximate analysis

Table 1 presents the elemental and proximate analysis of the raw materials employed in the formulation of a plastic char composite and pristine biochar. Table 1 indicates that rice straw possesses a carbon level of around 37%, whereas LDPE exhibits a significantly higher carbon content of around 83%. The total carbon content of the plastic char composite was 74%, whereas that of the biochar was 64.44%. The nitrogen and hydrogen percentages of the plastic char composite were around 2.59% and 4.84%, respectively. The incorporation of single-use plastic trash with rice straw to develop the plastic-char composite augmented its carbon content by 12.91%. The oxygen content in rice straw was elevated, whereas that in LDPE was significantly decreased. The plastic-char composite derived from these basic materials contains approximately 18% oxygen. The pristine biochar exhibited greater oxygen content (30.36%). The amalgamation of rice straw waste with plastic waste before the pyrolysis process resulted in a 40.71% reduction in oxygen concentration. The nitrogen and hydrogen levels in the plastic-char composite exceeded those of the pristine biochar. The H/C ratio was elevated in the plastic-char composite relative to pristine biochar, signifying that the stability of the plastic-char composite was inferior to that of pristine biochar. The O/C ratio in the plastic-char composite (<0.2) was lower than that of pristine biochar.

Similarly, the volatile matter concentration of rice straw was approximately 54%, but the LPDE had a significantly higher volatile matter content. However, the overall volatile matter in the plastic-char composite was about 64%. The plastic-char composite exhibited more ash and fixed carbon than the primary ingredients. Furthermore, the H/C was higher, and the O/C ratio was lower in the plastic-char composite than in the pristine biochar. This indicated that the plastic-char composite possesses slightly lower stability than the pristine biochar. However, a lower O/C ratio indicated that the plastic-char composite possesses hydrophobic characteristics. The rise in carbon content of pristine biochar and plastic-char composite, relative to the raw materials, is attributable to the pyrolysis process, which thermally concentrates carbon by facilitating the release of volatile and hydrocarbon fractions during carbonisation under anoxic conditions^{19,25}. The plastic-char composite showed higher carbon and lower volatile matter content compared to pristine biochar due to high carbon concentration and low ash content in single-use plastic waste, as well as synergistic carbonization reactions during co-pyrolysis²⁶. The low oxygen content in the plastic-char composite compared to pristine biochar could be ascribed to the low or no oxygen content in the plastic waste, which during co-pyrolysis, undergoes thermal cracking and dehydrogenation, generating aromatic structures and carbonaceous residues with low oxygen content²⁶. Low ash content in the plastic-char composite compared to pristine biochar could be ascribed to the same reasons as discussed above. A similar trend was reported in previous studies focused on producing plastic and carbon composites using biomass and plastic waste^{26–28}.

BET surface area analysis

The BET surface area analysis (Table 2) indicated a three-fold increase in the surface area of the plastic-char composite relative to pristine biochar. The total pore volume and average pore diameter of the plastic-char composite decreased by 35.71% and 73.27%, respectively. This reduction in pore volume and average pore diameter compared to pristine biochar can be attributed to pore blockage and structural compaction during the co-pyrolysis process, where thermal decomposition of plastic waste infiltrated the macro and micro pores of biochar, leading to pore blockage and a decrease in pore diameter. The surface area of plastic-char composite exhibits a three-fold increase relative to pristine biochar, which can be ascribed to the synergistic interaction between plastic volatiles and biomass. This interaction facilitates the development of a more porous carbon network, the filling of pores, and consequently, the prevention of pore collapse, thereby enhancing the exposure of the outer surface area. A similar trend was reported in earlier studies^{29,30}. Particle distributions and densities of the RSB and PRSB samples were analyzed for their suitability for cement applications, and the results are shown in Table 2. The particle distribution of the RSB is relatively coarser, having a particle distribution range

of 90–120 μm and 280–340 μm in D50 and D90, respectively, indicating the presence of large heterogeneous char particles. On the other hand, the PRSB has a finer particle distribution with D50 ranges of 70–80 μm and 150–180 μm for D90, inferring better fragmentation obtained from the process of co-pyrolysis and milling. The actual densities also served to further differentiate the materials, with the RSB having an actual density of $0.72 \pm 0.05 \text{ g cm}^{-3}$, as a result of the highly porous carbonaceous composition, and the actual density for PRSB is higher at $0.86 \pm 0.04 \text{ g cm}^{-3}$, owing to the partial densification during the co-pyrolysis process, making it less different in density from the cement.

XRD and FTIR analysis of rice straw biochar and plastic-char composite

Figure 1 shows the XRD analysis of pristine biochar and plastic-char composite. Figure 1a shows the XRD analysis graph of pristine biochar. It can be observed from the figure that peaks occur at 26° , 29° , 42° , 50° , and 67° (2θ angles). Similarly, Fig. 1b shows an XRD examination of the plastic-char composite, revealing peaks at 26° , 29° , 40° , 50° , and 67° (2θ angles). The broad and steep peaks at 26 and 29 degrees are caused by the existence of the (002) plane, which is associated with the presence of many layers of graphene and graphitic structure in the lattice plane with interlayer d-spacings of 0.175 nm and 0.158 nm, respectively. At a 2θ angle of 42 degrees, the lattice plane (100) indicates the presence of a hexagonally packed carbon structure perpendicular to the graphene layers, with a d-spacing of 0.119 nm. The lattice plane at 2θ angles of 50 and 67 degrees is caused by planes (110) and (222), which are linked to the presence of calcium oxide and hydroxide minerals in the plastic-char composite lattice. The plastic-char composite showed a higher intensity at the [002] peak than pristine biochar, indicating better graphitization during co-pyrolysis. Besides, the peak [100] at 42 degrees showed a higher intensity in the case of plastic-char composite than pristine biochar, indicating increased formation of the hexagonally packed structure compared to pristine biochar. The reason for the stronger [002] peak by plastic-char composite compared to pristine biochar can be attributed to the fact that the plastic decomposition during the co-pyrolysis process, along with rice straw waste, might have promoted the graphitic ordering and better stacking and alignment, resulting in high carbon content and improved crystallinity.

The FTIR analysis, presented in Fig. 2a and b, reveals the surface functional groups of the pristine biochar and plastic char composite. It can be observed from Fig. 3a that the pristine biochar contains various functional groups within the vibrational frequency range of 4000 to 500 (cm^{-1}). Specifically, the vibrational frequency at 3660 cm^{-1} is due to phenolic and hydroxyl functional groups, and the vibrational frequency range between 2600 and 2800 cm^{-1} indicates the presence of C-H aliphatic stretch³¹. The vibrational frequencies at 1430 cm^{-1} and 900 cm^{-1} indicate the presence of C=C aromatic stretch and $\text{C}=\text{CH}_2$ vinyl or aromatic alkene functional groups, respectively³². The vibrational frequencies at 750 cm^{-1} and 566 cm^{-1} indicate the presence of C—H aromatic and Si—O aromatic functional groups on the pristine biochar^{33–35}. The FTIR analysis of the plastic-char composite, as shown in Fig. 3b, showed the presence of various surface functional groups within the vibrational frequency range of 4000–500 cm^{-1} . The vibrational frequency between 2600 and 3600 cm^{-1} indicates the presence of hydroxyl and aliphatic C-H groups. Additionally, the plastic-char composite exhibited vibrational frequencies at 2600–2700 cm^{-1} , 2200 cm^{-1} , and 1700–2100 cm^{-1} , corresponding to the vibrational frequencies of the aldehyde group, alkene/nitrile, and carbonyl groups, respectively^{32,36}. The vibrational frequencies at $1475\text{--}1575 \text{ cm}^{-1}$ and 1075 cm^{-1} correspond to C=C aromatic/carboxylic group and C-O-C ester groups, respectively³⁷. Lastly, the vibrational frequencies between 550 and 770 cm^{-1} indicate the presence of C—H aromatic and Si—O aromatic functional groups^{33–35}. Moreover, a comparison of Fig. 3a and b reveals that the integration of plastic waste with rice straw during the co-pyrolysis process enhanced the prevalence of diverse surface functional groups, including aldehyde, ketonic/carbonyl, ester, and alkene/nitrile groups, on the plastic-char composite in contrast to pristine biochar. The augmentation of surface functional groups on the

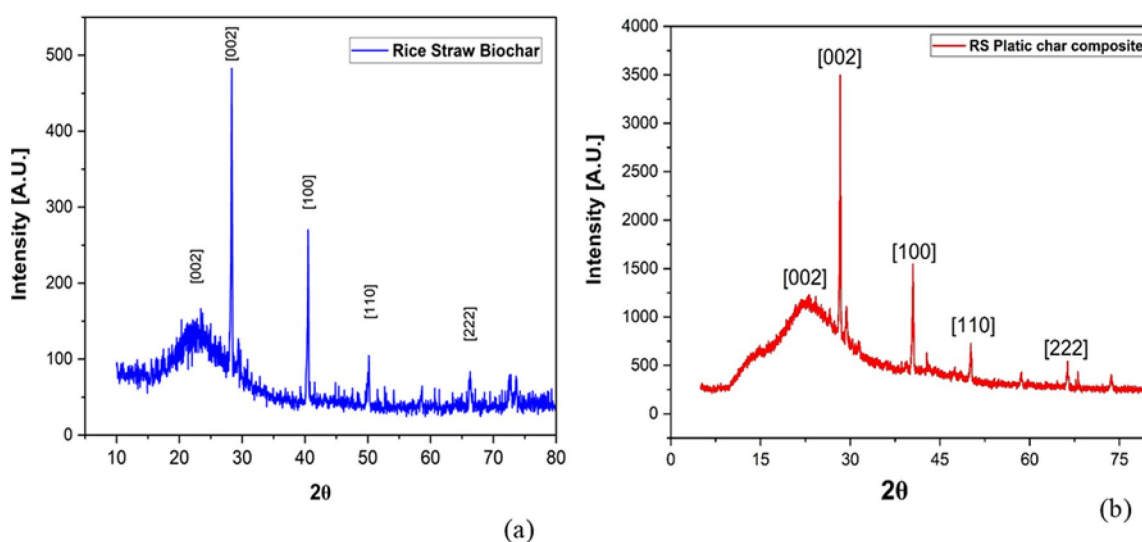


Fig. 1. (a) XRD analysis graph of rice straw biochar, (b) XRD analysis of plastic-char composite.

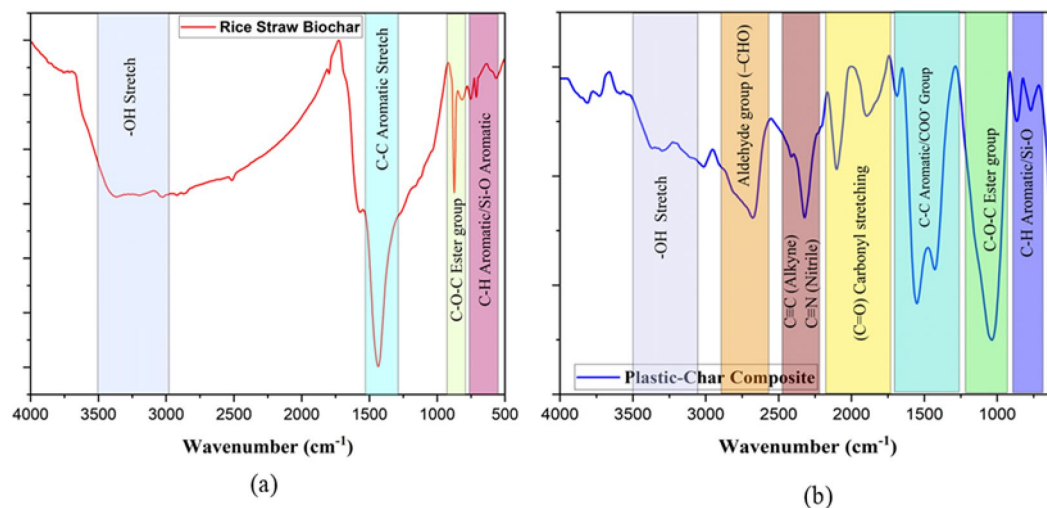


Fig. 2. (a) FTIR analysis of rice straw-derived pristine biochar, (b) FTIR analysis of the Plastic-char composite.

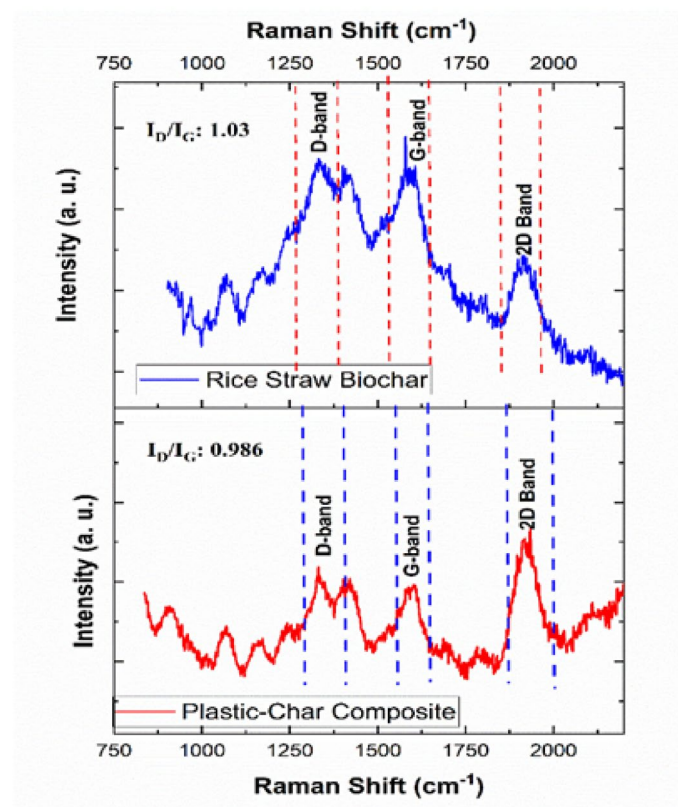


Fig. 3. Raman analysis of pristine biochar (RSB) and the plastic-char composite (PRSB).

plastic-char composite, in contrast to pure biochar, can be attributed to the thermal degradation and interaction of deteriorated plastic materials with biomass waste during co-pyrolysis. Furthermore, the co-pyrolysis of plastic waste with biomass waste results in radical species from plastic degradation interacting with the biochar matrix, leading to the formation of novel functional groups, including COOH, C=O, and C-N/C=N, on the surface of the biochar. A similar trend was observed in the previous studies, where it was reported that the co-pyrolysis of biomass waste with plastic waste improved the diversity of surface functional groups^{29,38}.

Raman and morphological analysis of RSB and PRSB

Raman analysis of biochar is done to analyze the carbon structure in biochar and specifically to evaluate its degree of graphitization, disorder, and aromaticity. Raman analysis of pristine biochar and plastic-char

composite is shown in Fig. 3. It can be observed from Fig. 4 that both RSB and PRSB exhibit characteristic peaks at wavelengths of 1350 cm^{-1} and 1580 cm^{-1} , corresponding to the D and G peaks, respectively³⁹. The G-peak at 1580 cm^{-1} represents the presence of graphitic sp^2 -hybridized carbon atoms in a hexagonal lattice, and the D-peak represents the presence of defective sp^2 -hybridized carbon rings or indicates disorder in the carbon structure⁴⁰. The infrared band region between 750 and 1200 cm^{-1} exhibited several peaks that correspond to the characteristics of in-plane and out-of-plane C–H bending of hydrogen atoms that adorn the edges of graphene sheets³⁹. The 2D peak position between 2600 and 2700 cm^{-1} indicates the presence of the second-order D-band corresponds to sp^2/sp^3 hybridized carbon structures. A highly graphitized carbon structure shows a graphitized carbon structure, whereas a weaker peak indicates the presence of an amorphous carbon structure⁴¹. The I_D/I_G in the Raman analysis of biochar indicates the presence of the degree of disorderness and graphitic carbon in the biochar. If I_D/I_G exceeds 1, it signifies a highly disordered amorphous carbon structure. A value of 1 suggests moderate disorder with an equal presence of both graphitic and amorphous carbon, while a value below 1 indicates a higher degree of graphitization⁴². Now, it can be observed from Fig. 3 that the I_D/I_G ratio in the case of RSB was greater than 1, indicating a higher degree of disorder and the presence of amorphous carbon structure. Whereas, in the case of PRSB, the I_D/I_G ratio was less than 1, indicating a higher degree of graphitization in PRSB compared to RSB. Further, the 2D peak between 2600 and 2700 cm^{-1} was more prominent in the PRSB case, indicating a highly graphitized carbon structure. Meanwhile, the RSB showed a weaker 2D peak than the PRSB, indicating the dominance of disordered carbon structure. Thus, it can be said that the co-pyrolysis of rice

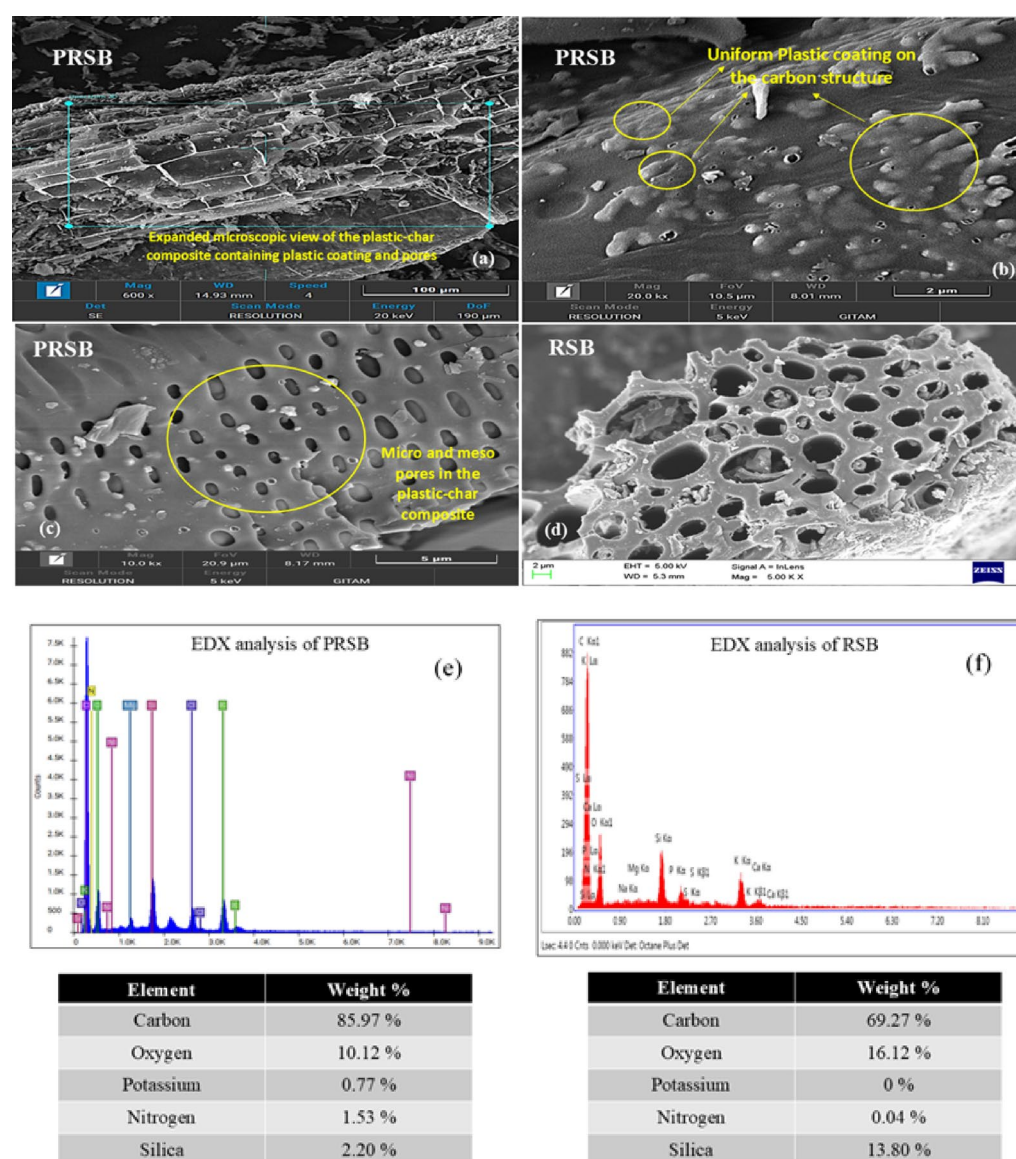


Fig. 4. (a–c): Morphological analysis of plastic-char composite (PRSB) captured between the scale of 2–100 microns at different magnifications, (d) morphological characteristics of rice straw biochar (RSB), (e) EDX analysis showing the elemental composition of PRSB, and (f) EDX analysis of RSB showing the elemental composition.

straw waste with single-use plastic enhanced the graphitization process, producing a more ordered and highly graphitized, dense carbon structure known as a plastic-char composite. The rise in graphitic structure observed in PRSB can be attributed to the thermal decomposition of plastic into dense carbon intermediates during the co-pyrolysis process, which may facilitate the formation of aromatic rings in the carbon structure, subsequently leading to condensation into a graphitic structure.

Figure 4 illustrates the morphological and EDX studies of PRSB and RSB. Figure 5 (a–c) demonstrates that the co-pyrolysis of plastic waste and rice straw waste yields composite materials characterised by micro and macro holes and a deposition of condensed carbonised residue from the plastic waste. Figure 4a, depicting an enlarged view of PRSB at 100 microns, clearly illustrates that the whole surface of the char composite is adorned with plastic-derived condensed carbon material. Figure 4b illustrates an enlarged depiction of a homogeneous layer of plastic-derived carbon material on the char's surface, resulting in the formation of the plastic-char composite. Moreover, Fig. 4d illustrates the SEM image of pristine biochar (RSB), indicating that the pyrolysis of rice straw waste produces a highly porous carbon material characterized by uniformly dispersed macro- and micro-holes throughout its structure. The EDX analysis of RSB, illustrated in Fig. 4f, indicates that the pristine biochar contains 69.27% carbon, with oxygen and silica comprising approximately 16.12% and 13.80%, respectively. The pristine biochar had a markedly reduced nitrogen concentration. In contrast, the PRSB EDX analysis (Fig. 4e) revealed a greater carbon content (85.97%) and nitrogen content (1.53%) than that of RSB. Moreover, the PRSB exhibited a reduction in oxygen and silica concentration relative to RSB. A decrease in oxygen and silica in the PRSB may be associated with the thermal and chemical reactions occurring during co-pyrolysis. A reduction in the oxygen content of PRSB can be attributed to the depletion of oxygen-containing functional groups in the plastic waste during the co-pyrolysis process, as the thermally unstable oxygenated groups, such as esters, alcohols, and carbonyls volatilize. A reduction in the silica concentration of PRSB may be attributed to the transformation of silica into volatile silicon compounds and the dilution impact resulting from an increase in carbon content. A previous study on the epoxy composites developed through the co-pyrolysis of plastic waste and natural resources reported similar morphological characteristics, where the authors found a uniform coating of thermally degraded plastic onto the surface of carbonized biomass⁴³.

High-resolution transmission electron microscope and SAED analysis of PRSB and RSB

High-resolution TEM analysis of the PRSB, as shown in Fig. 5a, revealed a non-uniform microstructure, in which the carbon-rich polymer particles are uniformly distributed within the carbon matrix. The lattice fringes indicate an amorphous character, with an interlayer spacing of approximately 0.34 nm, corresponding to the (002) planes of graphitic carbon. This confirms the presence of nanocrystalline graphitic regions embedded in the composite. The SAED analysis (Fig. 5b) of the PRSB showed concentric diffraction rings, confirming a polycrystalline arrangement of randomly oriented graphitic nanocrystals. Additional scattered spots along the rings suggest the presence of heteroatoms or small clusters, introducing localized lattice distortions. The well-defined diffraction rings and heteroatom-related spots indicate successful incorporation of heteroatoms or polymers within a predominantly crystalline graphitic framework, thereby improving chemical reactivity.

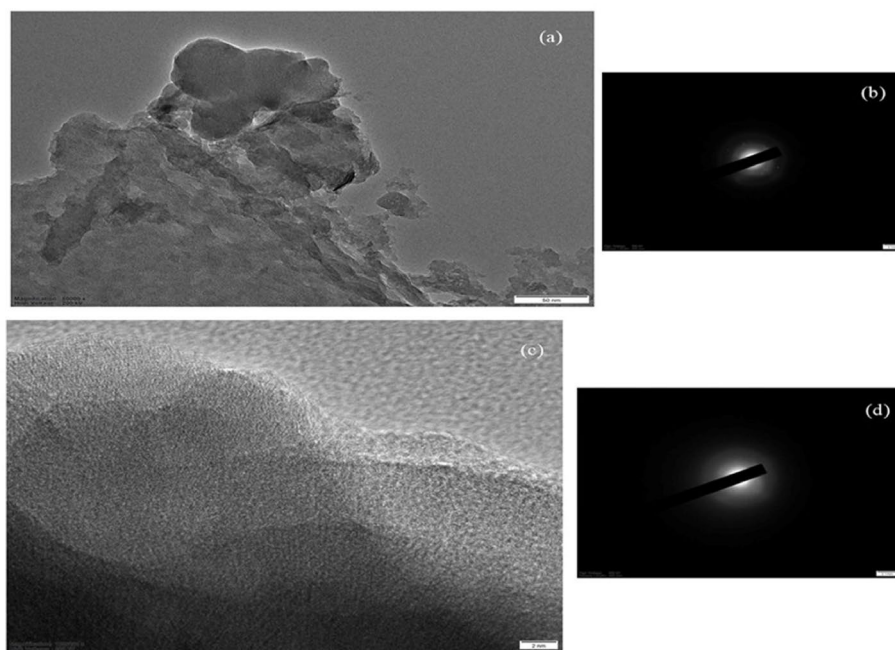


Fig. 5. (a) TEM analysis of PRSB, (b) SAED pattern analysis of PRSB, (c) TEM analysis of RSB, and (d) SAED pattern analysis of RSB.

Occasional lattice defects, such as edge dislocations, are visible in the fringes, while the surrounding polymer matrix exhibits interfacial bonding with the char regions.

High-resolution TEM images of rice straw biochar, as shown in Fig. 5c, show a predominantly porous and disordered carbon framework containing both amorphous regions and short-range graphitic domains. Lattice fringes associated with the graphite (002) planes (~ 0.34 nm) are observed intermittently and lack continuity, indicating incomplete graphitization and turbostratic stacking of the carbon layers. The corresponding SAED pattern (Fig. 5d) exhibits diffuse concentric rings, typical of turbostratically distributed carbon, composed of numerous small crystallites with random orientations. The overlapping nature of these rings further confirms turbostratic stacking and smaller crystallite dimensions compared to the plastic-char composite. Unlike the composite, no distinct heteroatom-related spots are visible, implying limited or irregular heteroatom incorporation. Structurally, the biochar contains abundant micro- and mesopores, as expected for chars derived from biomass, while its imperfect graphitic organization contributes to a high surface area. Together, the HRTEM and SAED results indicate that the PRSB has developed a more crystalline graphitic structure with evident heteroatom incorporation. In contrast, the RSB exhibited a predominantly disordered, turbostratic carbon matrix, characterized by smaller, less ordered nanocrystals, as indicated by the overlapping diffraction rings.

TCLP test of RSB and PRSB

The TCLP analysis of RSB and PRSB leachate showed remarkably low metal concentrations, with all metals well below their EPA regulatory limits (Table S3, Supplementary materials), confirming their environmental safety for use in soil and concrete applications. All the measured leachates of regulated metals (As, Cd, Cr, Pb, Se, Ba) were orders of magnitude below thresholds, with RSB generally showing marginally lower values compared to PRSB, presumably due to better metal stabilization imparted by its synthesis modifications. Concentrations of priority metals were also well below EPA TCLP guidelines (e.g., Pb < 3% of 5.0 mg L^{-1} ; Ba < 3% of 100 mg L^{-1}), indicating the absence of mobilization potential under disposal or concrete immersion in acidic conditions (pH ~ 5). The reduced leachability of RSB (32% less Pb, 33% less Ba compared with PRSB) might be related to RSB's doped structure, facilitating adsorption sites for increased metal adsorption over mobilization in basic cements (pH 12–13)⁴⁴. Unregulated metals (Ni, Zn, Cu) were present at Trace levels ($< 1 \text{ mg L}^{-1}$), as for most biochars produced from biomaterials, with no ecotoxicity potential for up to 10–20 wt% partial substitution of cements. These results align with prior studies on agricultural biochars, supporting RSB/PRSB as sustainable, non-hazardous pozzolanic additives that reduce cement demand while mitigating heavy metal risks in concrete.

Effect of mechanical properties

Effect of partial replacement of RSB and PRSB on compressive strength

The mechanical testing procedure and instruments used to evaluate the mechanical properties of partially replaced concrete specimens are shown in Fig. 6 (a–d). Figure 6a and b show the evaluation of compressive and flexural strengths of test specimens using the compressive strength testing machine and Universal Testing Machine, respectively. Figure 6c and d illustrate the procedure for evaluating the split tensile strength and permeability tests of the partially replaced concrete specimens.



Fig. 6. (a) Evaluation of compressive strength using the compressive strength testing machine, (b) Experimental procedure to evaluate flexural strength of the partially replaced concrete specimen using the Universal Testing Machine (UTM), (c) Experimental testing to evaluate split tensile strength, and (d) experimental testing procedure for permeability test.

Figure 7 illustrates the outcomes of the compressive strength test conducted on concrete cubes partially substituted with RSB and PRSB across curing durations of 3, 7, and 28 days. Figure 7 shows that across all three curing durations (3, 7, and 28 days), concrete partially replaced with both RSB and PRSB exhibited markedly superior compressive strength compared to standard concrete. When partially substituting cement with RSB, a 1% (w/w) increase in the replacement ratio enhanced the compressive strength of concrete cured for 3, 7, and 28 days by 21.42%, 13.79%, and 9.30%, respectively. Nonetheless, an additional increase in the replacement percentage beyond 1% and up to 7% (w/w) significantly reduced the compressive strength after 3, 7, and 28 days of curing, suggesting that a 1% (w/w) replacement is optimal for achieving the maximum compressive strength with pristine biochar (RSB). Similarly, the partial replacement of cement with plastic-char composite (PRSB) at 1% (w/w) significantly enhanced the compressive strength after curing for 3, 7, and 28 days by 38.88%, 32.43%, and 15.21%, respectively. A further increase in the replacement percentage to 3% (w/w) resulted in 42.11%, 35.89%, and 17.02% increases in compressive strength at 3, 7, and 28 days of the curing period, respectively. However, a further increase in the replacement percentage from 3% to 7% (w/w) resulted in a substantial decrease in compressive strength compared to 1% and 3%. However, unlike RSB, it remained higher than the control (Fig. 7). A similar increase in the compressive strength of partially replaced concrete structures using different feedstock-derived biochars was reported in earlier studies^{19,45}. Compressive strength enhancement in concrete partially substituted with RSB and PRSB can be attributed to the filler effect and pozzolanic reactions of RSB and PRSB during hydration¹⁹. The micro RSB and PRSB particles may have occupied the spaces created in the concrete structure after curing, resulting in denser packing. The existence of micropores in the RSB and PRSB enhanced internal curing through elevated water absorption, thereby sustaining a superior level of hydration. Furthermore, the RSB and PRSB, which contain silica and oxygen, may have interacted with calcium hydroxide released during cement hydration to produce additional calcium silicate hydrate (C-S-H), the primary component responsible for the strength of concrete¹⁸. The carbon content in the RSB and PRSB also plays an important role in enhancing the compressive strength. Carbon in biochar acts as a nucleation site for the growth of the hydration product and improves bonding at the interfacial transition zone, which is a critical zone where crack often initiates between aggregates and paste. Moreover, high carbon content also improves thermal and chemical stability, thus promoting long-term durability. However, an excess of RSB and PRSB in concrete, as observed in Fig. 7, reduces its compressive strength due to the dilution effect and weak bonding between the aggregate and cement, resulting from reduced chemical interaction¹⁸. The two-way ANOVA results (Table S4, supplementary material) confirm that compressive strength is significantly influenced by both concrete type (RSB vs. PRSB) and cement replacement level ($p < 0.001$). The higher F-value for replacement level indicates that strength development is predominantly influenced by dosage, underscoring the importance of optimal filler interactions and improved cement hydration at optimal substitution levels. Moreover, the significant interaction effect indicates that the influence of PRSB on compressive strength improvement is highly dependent on dosage, thereby justifying the non-linear strength development. The low error term variance confirms the reliability of the experimental results.

Effect on split tensile strength, flexural strength, and permeability

The split tensile and flexural strengths are critical properties of concrete that elucidate its behaviour and potential failure under tensile stress. The split tensile strength assesses indirect tensile strength under lateral compressive loads, while the flexural strength evaluates direct tensile strength during the bending of a concrete beam. Figure 8 (a-d) illustrates the split tensile strength and flexural strength of concrete with RSB and PRSB substitutes. It can be observed from Fig. 8a and b that the split tensile strength of concrete replaced with RSB and PRSB is higher

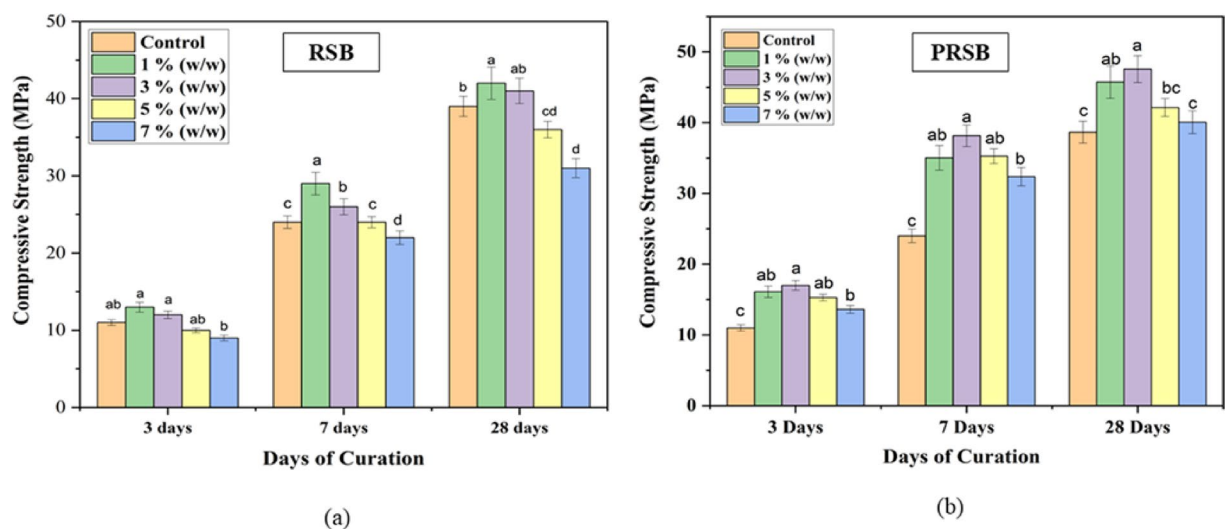


Fig. 7. (a) Compressive strength of M30 grade concrete partially replaced using RSB (1–7% w/w), (b) Compressive strength of M30 grade concrete partially replaced using PRSB (1–7% w/w) (number of replicates, $n = 3$ and confidence interval = 95%).

than that of plain concrete at all replacement percentages (1%, 3%, 7%, and 10% (w/w)). Furthermore, the split tensile strength of RSB- and PRSB-replaced concrete substantially increased with increasing replacement percentage to 1% (w/w) compared to the control, then gradually decreased as the replacement percentage increased to 7% (w/w).

Moreover, cement replacement with PRSB up to 1% (w/w) increased the split tensile strength by 46.15%, whereas replacement with RSB up to 1% (w/w) increased tensile strength by 39.13%. Similarly, the flexural strength of both RSB and PRSB-replaced concrete (up to 1% w/w) was increased by 17.24% and 20.87%, respectively, compared to the control.

The two-way ANOVA analysis further statistically validated the effects of curing time and cement replacement percentages on split tensile and flexural strength. The two-way ANOVA (Table S5) results confirmed that the curing age (A), the level of cement replacement (B), and the interaction between the two ($A \times B$) significantly affected the split tensile strength (STS), as evidenced by the extremely high F-values and p-values of less than 0.001. The total variation in split tensile strength was 43.59, of which 28.64 (approximately 65.7%) was accounted for by the curing age alone. The F-value of 521.5 also supports that the difference in influence between curing age and cement replacement level was quite large. Compared to this, the cement replacement level accounted for 9.87 ($\approx 22.6\%$) of the total variation, with an F-value of 90.1, which, although highly significant, was much lower than that of curing age. This implies that material substitution impacted the tensile strength, but its effect was secondary to the hydration time. The interaction term $A \times B$ accounted for 4.26 ($\approx 9.8\%$) of the variance and was statistically significant ($F = 19.4, p < 0.001$). This suggests that the impact of the level of cement replacement was not constant over the curing ages. Rather, the impact of the percentage of replacement varied over time, supporting the coupled hydration and microstructure evolution process. For flexural strength, the two-way ANOVA analysis (Table S6) revealed that curing age (A) exerted the strongest influence on split tensile strength, contributing 28.64 of the total sum of squares ($\approx 65.7\%$) with an extremely high F-value (521.5, $p < 0.001$), indicating dominant time-dependent hydration effects. Replacement level (B) was also significant ($F = 90.1, p < 0.001$), accounting for $\approx 22.6\%$ of the variation, confirming that cement substitution alters tensile performance. The significant $A \times B$ interaction ($F = 19.4, p < 0.001$) suggests that the effect of replacement level depends on curing duration. The low error variance ($MS = 0.027$) demonstrates strong experimental reliability and consistent tensile strength development trends.

Overall increase in the split tensile and flexural strength of RSB and PRSB substituted concrete compared to the control can be attributed to mechanisms such as pore filling by RSB and PRSB particles, delay in crack initiation and propagation due to biochar's ability to arrest microcracks, and better internal hydration due to water retention by biochar particles^{46,47}. The enhanced split tensile strength and flexural strength of PRSB-replaced concrete, in comparison to RSB-replaced concrete, may be ascribed to the physicochemical features of PRSB. The elevated carbon content, characterised by a more aromatic structure, together with the incorporation of PRSB in concrete, led to superior mechanical integrity of carbon particles, greater stress transmission within the concrete matrix, and higher resistance to micro-cracking. The PRSB produced a more uniform particle shape with polymer coating, rendering it appropriate for filling micropores, decreasing porosity, and enhancing packing density. The plethora of surface functional groups, including phenolic, aldehyde, carbonyl, C=C

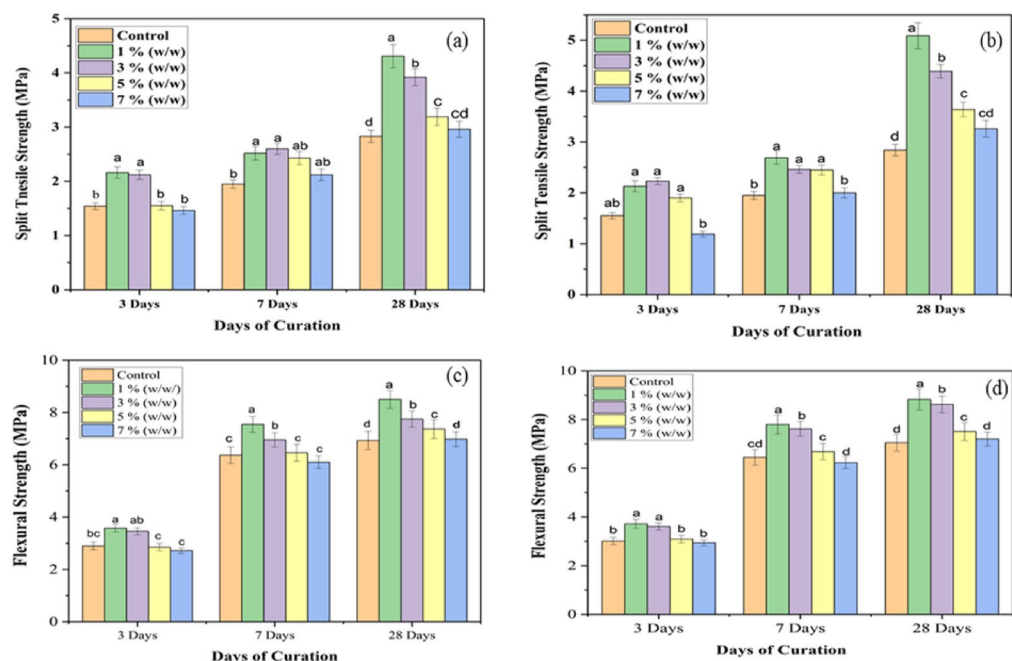


Fig. 8. (a) Split tensile strength of RSB-replaced concrete, (b) split tensile strength of PRSB-replaced concrete, (c) Flexural strength of RSB-substituted concrete, (d) flexural strength of PRSB-substituted concrete (number of replicates, $n = 3$ and confidence interval = 95%).

S. No.	Biochar type	Pyrolysis condition	Cement replacement percentage	Compressive strength (MPa) (28 days)	Flexural strength (MPa) (28 days)	Split tensile strength (MPa) (28 days)	References
1	Rice husk biochar	Temp: 500 °C	1	27.87	7.21	–	19
			2	26.67	7.39		
			4	27.24	7.16		
2	Forest wood biochar	Temp: 500 °C	1	27.53	7.04	–	19
			2	26.45	6.83		
			4	25.89	6.61		
3	Rice husk biochar	Temp: 600 °C	5	36.8	–	3.38	48
			10	38.9		3.47	
			15	37.3		3.58	
			20	35.7		3.5	
			25	33.9		3.3	
4	Eucalyptus wood biochar	Temp: 500 °C	0.65	11.7	–	1.7	45
			3.2	11.4		2.0	
			6.5	12.6		1.8	
			9.5	8.0		1.6	
			13.5	6.90		1.9	
5	Rice husk biochar	Temp: 500 °C	0.25	22.3	4.4	2.8	49
			0.5	32.1	4.3	3.8	
			0.75	28.6	4.1	3.1	
			1	29.7	4.8	3.2	
6	Paper waste biochar	Temp: 500 °C	0.1	38	4.4	3.6	49
			0.5	34	3.4	2.72	
			1	30.5	3.8	2.83	
7	Carya cathayensis peel biochar	Temp: 500 °C	1	45.8	–	3.04	50
			2	49.73		3.21	
			3	42.90		2.98	
			4	39.97		2.91	
			5	35.67		2.59	
8	Rice Husk biochar	Temp: 700 °C	5	27.09	–	–	51
9	Rice straw biochar	Temp: 800 °C	1	44	8.4	4.4	This study
			3	43	7.8	3.9	
			5	38	6.6	3.3	
			7	33	6.5	3.1	
10	Platic-char composite	Temp: 800 °C	1	45	8.6	5.0	This study
			3	47	8.4	4.4	
			5	41	6.8	3.8	
			7	39	6.5	3.4	

Table 3. Comparison of Split tensile and flexural strength with previous studies.

aromatic, ester, and silica oxide groups in the PRSB, may have enhanced the interfacial bonding with hydration products (C-S-H) and the strength of the interfacial transition zone. The superior structural integrity and customised porosity with graphitic domains in the PRSB may have improved the load distribution capacity of the concrete, leading to increased split tensile and flexural strength. Table 3 presents data obtained from prior investigations in which biochar served as a cement substitute in mortar and concrete at various replacement percentages, juxtaposed with data generated in the current investigation. It is evident that the use of PRSB as a 1% (w/w) cement replacement material exhibits superior performance in terms of compressive, flexural, and split tensile strength compared with previously documented investigations. Nonetheless, no direct comparison of PRSB with any previously published studies involving plastic/biochar-based composites has been conducted, as this material is unique and has not been documented in prior research.

Effect on permeability coefficient

Figure 9 presents the permeability coefficient of PRSB- and RSB-incorporated concretes in comparison with plain concrete after 28 days of curing, clearly demonstrating a systematic reduction in water transport with biochar incorporation. Compared to the control mix, RSB-concrete and PRSB-concrete exhibit lower permeability coefficients, thereby attesting to the effectiveness of biochar as an additive in altering pore structure. Worth stating, too, is that for every substitution level ranging from 1% to 7% by total weight, PRSB performs

Mix design code	Cement (kg/m ³)	Plastic-char composite (kg/m ³)	Fine aggregate (kg/m ³)	Coarse aggregate (kg/m ³)	W/C ratio (%)
M0	400	0	665	1170	0.45
M1	396	4	665	1170	0.45
M3	388	12	665	1170	0.45
M5	380	20	665	1170	0.45
M7	472	28	665	1170	0.45

Table 3. Mix design parameters and quantities to cast concrete cubes and beams.

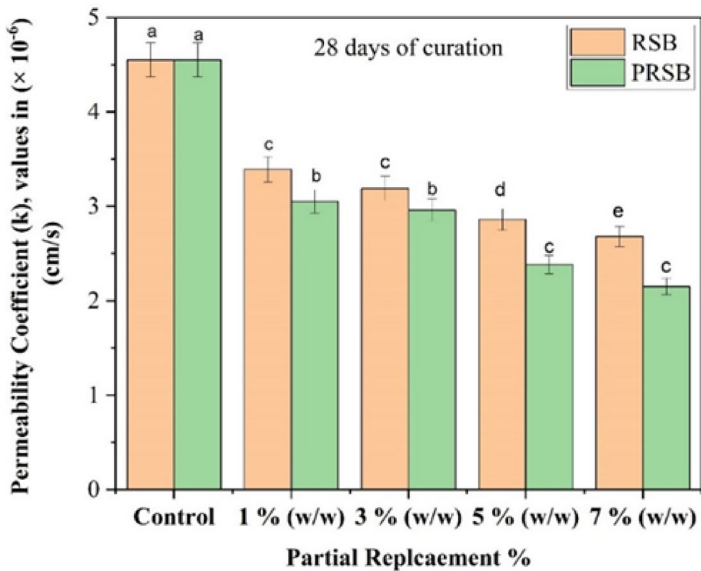


Fig. 9. Variations in the permeability coefficient of RSB and PRSB substituted concrete cured for 28 days (number of replicates, *n* = 3 and confidence interval = 95%).

better than RSB. Two-way ANOVA results (Table S7) show that both concrete type and replacement level had a significant effect on the response variable ($p < 0.001$). The F-value of replacement level (71.8) was higher than that of concrete type (48.6), which clearly indicates that it had a more dominant effect on the total variability. The sum of squares of replacement level (18.95) was almost three times higher than that of concrete type (6.42). The interaction effect was also significant ($F = 7.3$), which clearly indicates that the effect of replacement level was not uniform on different concrete types. The low error mean square (0.066) indicates good experimental precision and reliable factor differentiation.

When the substitution level is raised to 1%, the permeability coefficient decreases by about 10%; when it is further increased to 3%, 5%, and 7%, it decreases by 10%, 16.78%, and 19.77%, respectively. This indicates that as the PRSB dosage increases, the discontinuity in the pores becomes more pronounced. This indicates that PRSB is more successful than RSB in blocking the connected capillary pathways. The lower permeability of PRSB may be due to its modified shape, hydrophobic aromatic functional groups, and higher bulk density.

These permeability trends are strongly corroborated by the sorptivity results, which show a parallel reduction in capillary water uptake with increasing RSB and PRSB content. As sorptivity is much more sensitive to the connectedness of capillary pore systems than to the porosity, the simultaneous reduction in permeability coefficient and sorptivity values provides additional evidence that the pore system itself has become more refined yet less connected. The coordination between the transport properties provides further evidence that biochar incorporation reduces the continuity of water-conducting channels in the cementitious matrix. A similar trend of decreased permeability coefficient and sorptivity after partial replacement of cement with biochar was observed in previous studies, due to decreased conductivity and a higher rate of hydration and pozzolanic reactions^{52,53}. Mechanistically speaking, the noted reductions in permeability and sorptivity can be ascribed to a variety of coupled mechanisms: (i) the physical occupation of microvoids by RSB and PRSB particles with a fine dispersion, (ii) pozzolanic reactions between the Reactive Biochar surface and calcium hydroxide, thereby augmenting calcium-silicate-hydrate formation, (iii) augmentation of interfacial transition zone quality and packing density, (iv) internal curing action associated with Biochar water adsorption and subsequent release, and (v) biologically inherent hydrophobic properties of RSB and PRSB that further suppress water entry. Underlying all of these phenomena is the augmentation of durability performance with PRSB having a more substantive benefit than RSB⁵⁴.

Optimized replacement percentage and biochar type for M30 grade concrete mix

The details and results of the optimized mechanical properties and replacement percentages are shown in Fig. 10. This work systematically investigated the optimization of the mechanical properties of M30 grade concrete by partially replacing cement with a plastic-char composite (PRSB) and rice straw biochar (RSB) using a Box-Behnken experimental design. The impact of two key variables—curing time and replacement percentage—on compressive strength was analyzed. Surface and contour plots (Fig. 10a and b) of the experimental data illustrate a distinct trend: the compressive strength of the concrete increases with both the duration of curing and the percentage of PRSB replacement, notably up to 3%. The observed trends result from the increased pozzolanic reactivity and enhanced pore-filling capabilities of biochar materials, which refine the microstructure and facilitate the formation of a denser interfacial transition zone within the concrete matrix. Previous studies have reported similar outcomes, indicating that incorporating biochar and plastic-derived additives enhances the mechanical integrity and durability of concrete at optimal replacement levels^{45,55}. Subsequent analysis using the Box-Behnken optimization model revealed that a maximum compressive strength of 56 MPa could be attained with a PRSB replacement level of 3% (w/w) after a 28-day curing period (Fig. 10m). This value exceeds the control mix, indicating that a well-optimized PRSB addition may improve hydration kinetics and the formation of calcium silicate hydrate (C-S-H) gel, which is necessary for strength development. The ideal replacement threshold aligns with the literature, demonstrating that excessive replacement may dilute the cementitious characteristics and reduce strength beyond its limitations. PRSB works well as a partial cement substitute in M30 concrete, enabling sustainable material use while maintaining structural integrity for practical applications. Plastic char and biochar synergistically enhance nucleation sites, refine porosity, and improve binder-aggregate interaction, thereby enhancing mechanical properties. Experimental results showed that RSB and PRSB substitution increased split tensile strength, notably up to 3% (w/w) and 28 days of curing. This improvement is attributed to biochar's pozzolanic properties, which increase the nucleation sites of calcium silicate hydrate (C-S-H), and the plastic-char particles' filler effect, which enhances the interfacial transition zone and reduces microcracks. The Box-Behnken optimization model ($R^2=0.99$) showed that a 3.8% (w/w) PRSB substitution and a 28-day curing time may achieve a split tensile strength of 4 MPa, outperforming the control mix. According to the literature, modest carbon-rich waste additions increase matrix densification, bond strength, and durability, whereas excessive substitution reduces the cementitious content and mechanical performance. These findings demonstrate that biochar and PRSB enhance tensile performance, enabling the manufacture of low-carbon concrete using agricultural and plastic waste. Figure 10 (i, k) shows the contour diagram of RSB and PRSB-replaced M30 concrete flexural strength with curing time (0–28 days) and replacement percentage (0–7%). The contour graphic indicates that RSB-substituted concrete exhibited a flexural strength of 7–8 MPa at a 2–5% (w/w) replacement ratio and after 25–28 days of curing. The contour diagram of the PRSB substitution indicates a flexural strength of 8–10 MPa, with a replacement percentage of 3–5%, and a curing duration of 25–28 days. When the replacement ratio was 1–3%, PRSB-substituted materials had flexural strengths of 10–12 MPa. RSB and PRSB-based concrete surface plots exhibited a three-dimensional parabolic curve with respect to replacement % and cure period. The flexural strength of RSB and PRSB-based concrete increased during

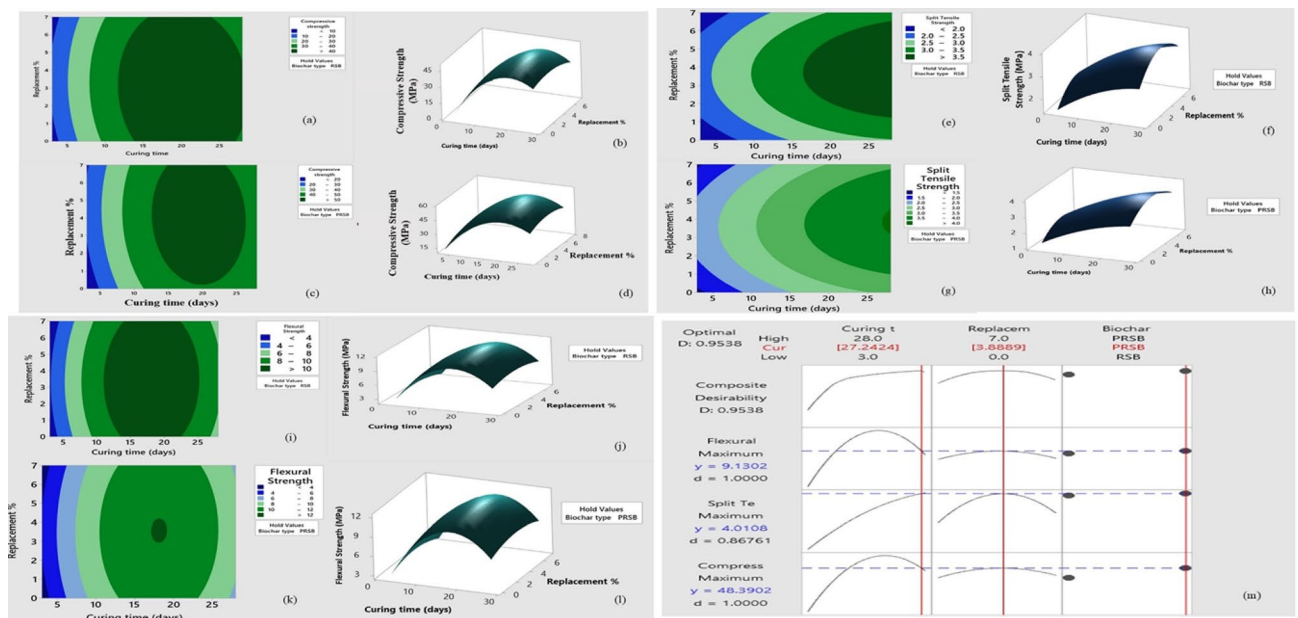


Fig. 10. (a,c): Contour diagram of compressive strength of RSB and PRSB substituted M30 concrete, (b,d): Surface plot of compressive strength with respect to replacement % and curing time, (e,g): Contour diagram of split tensile strength of RSB and PRSB substituted M30 concrete, (f,h): surface plot of split tensile strength, (i,k): Contour diagram of flexural strength of RSB and PRSB substituted M30 concrete, (j,l): surface plot of flexural strength, and (m) optimized response diagram of compressive, split tensile, and flexural strength.

curing, reaching a peak at a 2–3% replacement rate. The Box–Behnken optimisation model ($R^2 = 0.98$) showed that a 3.8% (w/w) PRSB replacement and 27–28 days of curing may produce a flexural strength of 9.13 MPa.

Thus, the final optimization model, as shown in Fig. 10m, indicates that the maximum value of compressive strength (48.39 MPa), split tensile strength (4.01 MPa), and flexural strength (9.13 MPa) can be achieved when the cement in the concrete mix is partially replaced with PRSB at 3.88% (w/w) and cured for 28 days.

Post-mechanical testing SEM and XRD analysis to understand the chemistry behind the fluctuations in the mechanical strength of plastic-char partially-replaced sustainable concrete

Figure S1 (Supplementary Materials) presents the XRD patterns of post-mechanically tested plastic-char-based concrete, clearly demonstrating how partial cement substitution governs both phase evolution and mechanical response. The control mix exhibits sharp and well-defined reflections corresponding to crystalline cement hydration products, particularly portlandite ($2\theta \approx 18^\circ, 29^\circ, 34.1^\circ, 46.1^\circ, 47.3^\circ, 51^\circ$, and 59.7°) and calcium silicate hydrate (C–S–H) around 29° , while the weak feature near 21° is indicative of poorly crystalline or semi-ordered hydration phases. Upon replacing cement with the plastic–rice–straw biochar (PRSB) composite at low dosages (1–3 wt%), these characteristic peaks remain pronounced, which, importantly, indicates that the introduction of finely divided char does not suppress hydration but rather facilitates it. This behavior is indicative of a heterogeneous nucleation mechanism, where char particles with a high surface area and chemical activity behave as natural sites, inducing C–S–H precipitation and consequently accelerating local hydration, which in turn facilitates densification of the matrix. Accordingly, even clinker dilution at these rates is somewhat counterbalanced by achieving better microstructural packing and more abundant continuity of load-bearing hydration products. This clearly accounts for the enhanced strength parameter (compressive strength) observed. By contrast, at higher replacements ($> 3\%$), the XRD patterns reveal continuous peak broadening and a loss of intensity, particularly in the case of portlandite and C–S–H-related reflections, along with the appearance of a broad amorphous hump that can be ascribed to the carbonaceous PRSB phase. This change indicates a shift from a hydration-precipitation-dominated crystalline skeleton to an amorphous framework, with the addition of more and more char, which dilutes hydrate phases and loses some spatial connectivity. As a result, the reinforcing and nucleating benefits of the char are outweighed by excessive clinker dilution, limiting further mechanical gains at 5% and 7% replacement levels, as also summarized in the phase parameters reported in Table S2. These phase-level interpretations are strongly corroborated by SEM observations (Figure S2, Supplementary Materials). At low replacement levels ($< 3\%$), PRSB and RSB particles are homogeneously distributed and strongly embedded in a dense, continuous C–S–H network, demonstrating their dual role as micro-fillers and hydration nucleation sites provided by the biochar particles. With this, the ITZ would become denser, leading to better stress transfer from carbonaceous inclusions to the cement paste, and thus crack initiation is delayed, with all related benefits in terms of load-bearing capacity in compression. In contrast, at higher levels of replacement, the microstructures become more heterogeneous, with larger voids, weaker interfacial bonding, and areas of incomplete hydration (including around char/carbon clusters) that serve as stress concentrations, inducing early microcracking^{21,56,57}.

From a sustainability perspective, the optimized low-level substitution of cement with waste-derived plastic–biochar composites achieves a synergistic balance between performance and environmental benefit. By simultaneously valorizing plastic and agricultural residues while reducing clinker content, the process lowers embodied carbon and diverts waste from landfills without compromising—and indeed initially enhancing—mechanical performance. This mechanistically driven optimization highlights that the sustainability advantage of char-modified concrete is maximized not at maximum replacement, but at an experimentally defined threshold where microstructural refinement, mechanical integrity, and carbon footprint reduction are achieved simultaneously.

Life cycle assessment of RSB and PRSB replaced concrete

The Life Cycle Assessment (LCA) of cement substitution in M30 grade concrete with rice straw biochar (RSB) and plastic-char composite (PRSB) reveals considerable environmental impacts, evaluated via the cradle-to-gate methodology and utilizing the RiCiPe Mid-point H approach (Fig. 11a and b). This technique objectively assesses various effect categories to determine the sustainability of integrating biochar composites as partial cement replacements at varying proportions—1%, 3%, 5%, and 7% (w/w). At a 1% substitution level (w/w), the LCA shows a significant decrease in several environmental impact metrics compared to conventional concrete. A significant reduction in global warming potential—from 80% to 70%—indicates a substantial reduction in the carbon footprint. Likewise, stratospheric ozone depletion decreases significantly from 55% to 30%, while fine particle emissions diminish from 70% to 50%. The consequences of freshwater toxicity and human carcinogenicity are significantly reduced to 65% and 50%, respectively, accompanied by decreases in land-use impact from 60% to 50%, fossil-fuel scarcity from 45% to 35%, and water consumption from 45% to 30%. The decrease in global warming potential can be linked to reduced cement use, carbon sequestration in the stable aromatic matrix of both chars (estimated at 870 kg CO₂/kg-eq/tonne feedstock avoided), avoidance of clinker calcination CO₂, and avoided emissions from open rice straw burning and plastic incineration, which release CH₄, N₂, and black carbon. One previously reported study found that LCA analysis showed waste rice straw biochar-based concrete reduced CO₂ emissions by 63% relative to OPC, attributed to both biochar's carbon-negative nature and improved CO₂ fixation efficiency⁵⁸. A decrease in the stratospheric ozone depletion contribution can be associated with the fact that controlled pyrolysis emits far fewer halogenated and reactive nitrogen precursors than open burning or fossil fuel pathways. Pyrolysis emits fewer chlorine, bromine, nitrogen, and VOC precursors than fossil-derived naphtha production, resulting in lower or equivalent acidification, ozone depletion, and smog formation potentials. With reference to an earlier study⁵⁹, Cradle-to-gate LCA has shown that ozone depletion, toxicity, and hazardous waste exhibit decreasing trends as biochar

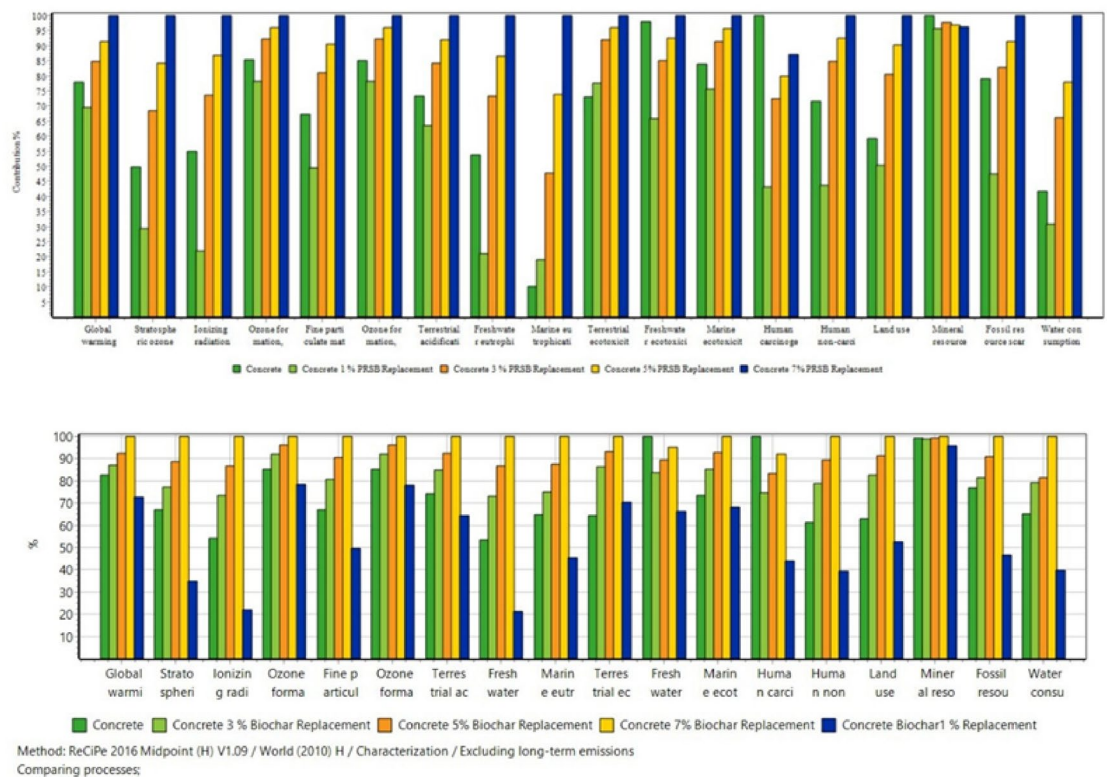


Fig. 11. (a) Life cycle assessment of PRSB-substituted concrete, (b) LCA of RSB-substituted concrete.

content in cement composites increases from 0 to 20 wt%. A significant decrease in the toxicity of freshwater and the carcinogenicity to humans can be attributed to reduced cement consumption and the addition of biochar to the cement matrix. In summary, the functional groups of biochar (carboxyl, hydroxyl, and carbonyl) bind trace heavy metals in the cement matrix, and the diversion of waste from fields decreases dissolved organic carbon. Reduced clinker content also lowers leachable heavy metal load (Cr, Pb, Mo). Further, cement production often involves leaching of heavy metals into water; plastic incineration generates dioxins, furans, and PAHs; and open straw burning produces benzene and formaldehyde. Both reductions in cement consumption and the recycling of agro and plastic waste may have reduced the load of these pollutants within the environment, thereby reducing freshwater toxicity and human carcinogenicity⁶⁰.

These enhancements highlight the ecological benefits of substituting a minimal amount of cement with biochar composites, emphasizing reductions in resource depletion and pollutant emissions. Nonetheless, when the replacement percentage increased from 1% to 3%, the environmental benefits plateaued, with effects approaching those of conventional concrete. This indicates a limit beyond which additional replacement fails to provide additional environmental benefits, likely due to a trade-off between the advantages of reducing cement and the environmental costs associated with manufacturing or using substitute materials. The LCA study reveals that increasing the 5% replacement level to 7% (w/w) results in a reversal, with environmental consequences significantly worse than those of standard concrete. This increase can be attributed to various factors, including the properties of materials that influence concrete performance, resource consumption or emissions associated with the higher composite content, and material processing and durability at higher replacement levels. For example, in one study, it was found that at higher replacement levels, biochar cementitious composite needed additional superplasticizer (SP) to ensure the mechanical strength and durability of concrete, which has a global warming potential of 1.5 kg CO₂-eq per kg of additive⁵⁵. The SP dosage grows non-linearly with biochar content, introducing increasing LCA burdens across GWP, freshwater toxicity, and fossil fuel scarcity categories. Further, higher doses of biochar in the cementitious metrics significantly increase water demand to maintain standard consistency, which may increase additional pressure on water resources if implemented at a larger scale⁶¹. Another study reported that biochar reportedly contains a variety of toxicants, including heavy metals, PAHs, polychlorinated dibenzodioxins, polychlorinated dibenzofurans, and VOCs — these can be present in the biomass or form during pyrolysis⁶². At 1%, the mass of these toxicants is negligible. At 5–7%, it is five to seven times higher, creating measurable contributions to human carcinogenicity and freshwater toxicity. Biochar production itself using small reactors may have adverse impacts, such as particulate emissions and air pollution, due to their inability to effectively utilize by-products. Pre-treatment (cleaning, drying, and grinding) can consume significant energy. At 5–7% replacement, the char mass required (~12–18 kg/m³) is 5–18× the 1% requirement, and all production-phase burdens scale accordingly⁶³.

The study reveals similar tendencies when comparing the two substitute materials: rice straw biochar and plastic-char composite. Both materials have minimal environmental effects at a 1% replacement rate. Although the overall impacts are comparable, specific categories such as ozone depletion, terrestrial and freshwater ecotoxicity, stratospheric ozone depletion, human carcinogenicity, and water consumption are generally greater for rice straw biochar than for the plastic-char composite at equivalent replacement rates. This variance may arise from discrepancies in production procedures, chemical composition, or lifetime emissions of the two compounds.

The LCA findings indicate that partial substitution of cement in M30 concrete with biochar composites at low levels (about 1% w/w) can markedly reduce environmental harm across all categories, particularly global warming potential and toxicological effects. Nevertheless, elevating replacement rates above moderate thresholds may undermine these advantages, underscoring the necessity of optimising substitution levels. The selection between rice straw biochar and plastic-char composite might have certain environmental implications, with plastic-char often exhibiting fewer adverse effects in some categories. These findings are crucial for developing sustainable concrete plans, offering avenues to mitigate the environmental impact of building materials while maintaining performance and promoting the use of waste-derived biochar composites. The cradle-to-gate life cycle assessment technique, coupled with thorough sensitivity analysis using the RiCiPe Mid-point H methodology, offers an extensive review to inform sustainable material selection and concrete mix formulation.

Further, the Global Warming Potential (GWP 500) impact analysis (Figure S3, supplementary material) indicates that substituting 1% (w/w) plastic-char composite (PRSB) for cement in concrete markedly reduces long-term global warming effects relative to conventional (plain) concrete. The GWP evaluation specifies: Global Warming Potential from Fossil Fuel Emissions: A drop from 80% to 72% in comparison to plain concrete, indicating a significant reduction in the contribution of fossil-derived greenhouse gases and Global Warming Potential from Biogenic Emissions: A more substantial reduction, from 78% to 60%, underscores the efficacy of 1% (w/w) PRSB incorporation in alleviating the effects of global warming associated with biogenic carbon sources. Nonetheless, augmenting the replacement percentage above 5% (w/w) inverts this tendency. At elevated substitution rates, there is a significant increase in the likelihood of global warming effects from fossil fuel burning, biogenic emissions, and land alteration, with values surpassing those of traditional concrete. This indicates a theoretical threshold, beyond which the environmental impact linked to increased quantities of substitute material (including processing, transportation, and potential effects on concrete durability) surpasses the advantages of cement reduction.

Furthermore, CO₂ uptake research indicates that even minor to moderate cement substitution (1% or above) significantly improves the long-term carbon sequestration capacity of concrete (as per the LCA analysis results): CO₂ Uptake: Ranges from 20% for plain concrete to 80% for concretes using PRSB, illustrating the significant carbon-absorbing potential facilitated by the inclusion of PRSB in the cementitious matrix. The noted decreases in GWP for both fossil and biogenic emissions at the 1% replacement threshold can be ascribed to several factors: reduced clinker utilization: PRSB directly substitutes a portion of the high-CO₂-emitting clinker component, thereby diminishing fossil-fuel-associated emissions. Biochar Characteristics: PRSB, a carbon-dense and durable substance, sequesters biogenic carbon and promotes additional CO₂ absorption during concrete curing and carbonation. Impact of Circular Economy: Utilizing agricultural waste (rice straw) for biochar synthesis leverages renewable biogenic carbon sources and promotes waste-to-resource strategies. The significant rise in CO₂ absorption (from 20% to 80%) in concrete containing PRSB demonstrates biochar's potential for application as carbon-neutral or carbon-negative building materials. However, the pattern reverses when the substitution ratio is above 5%. The significant increase in GWP from all emissions and land use change may result from: heightened energy and emissions associated with the production, processing, and transportation of larger volumes of PRSB, possible deterioration in concrete performance resulting in increased resource consumption or reduced service life, and environmental trade-offs related to land conversion for the large-scale sourcing or processing of biochar feedstocks. These findings highlight the necessity of optimizing the replacement rate: whereas little additions (<5%) of PRSB to concrete provide significant long-term climatic advantages, excessive substitution may impose additional environmental costs, undermining the benefits. The significant and beneficial effect on CO₂ absorption at substitution rates of 1% and above further reinforces the argument for the restricted replacement of cement with PRSB as a judicious strategy for climate-conscious concrete design.

Further, Figure S4 (supplementary materials) shows the environmental effect contributions of Plain Concrete, 1% Partially Replaced PRSB-based Concrete, and 1% Partially Replaced RSB-based concrete in several categories (including transportation, electricity uses, cement production, plastic-char composite, and rice straw biochar production) using life cycle assessment (LCA). The figure indicates that replacing cement with biochar (either PRSB or RSB) at 1% consistently lowers the relative contribution of cement to most environmental effect categories when compared to ordinary concrete. Cement remains the most important contributor across most categories, but its share decreased when PRSB or RSB was used for partial replacement, notably in areas such as global warming potential, ionising radiation, and terrestrial acidification. Cement manufacture has a big impact on the environment when it comes to plain concrete, especially when it comes to global warming, ozone creation, terrestrial acidification, eutrophication, human carcinogenic toxicity, and the depletion of mineral or fossil resources. In the 1% PRSB-based concrete, the amount of cement that contributes to each effect category is obviously lower, while the amount of biochar production and transport contribution were higher. This is because more materials were included, which made the environmental load a little more spread out. The 1% RSB-based concrete showed patterns that are comparable to those of PRSB. It is possible to reduce the environmental impact of cement, but this will cause the agricultural byproduct (rice straw biochar) supply chain and power processing to experience a small increase in their own impacts. The most important environmental advantage of partially replacing using PRSB or RSB is that it consistently lowers the environmental effects of cement. Cement is the most carbon- and energy-intensive ingredient of concrete. There are certain trade-offs, such as technology

and transportation having a little bit more of an effect, but these effects are still rather small. In general, the results demonstrated that both RSB and PRSB substitution help the transition to lower-impact concretes by employing agricultural or plastic waste streams, which is in line with the concepts of a circular economy. Even a small amount of tailored biochars replacing cement may make a big difference in LCA, making it a tool to cut emissions and reduce resource usage in the building industry. These results clearly support the idea that biochar-concrete mixes are good for the environment and for using resources efficiently, which is in line with the goals of materials science to reduce carbon emissions and turn waste into value.

Sensitivity analysis

The LCA sensitivity analysis for both Rice Straw Biochar (RSB) and Plastic Modified Rice Straw Biochar (PRSB) used as partial substitutes for cement (Figure S5) shows how energy supply and transportation affect the results systematically. The midpoint normalized representative indicators: GWP, Acidification, Eutrophication, Particulate Matter Formation, and Photochemical Oxidant Formation indicate an apparent reliance on the energy supply in the electricity sector for the production of RSB and PRSB, specifically for the energy required in the pyrolysis process. A clear downward shift in impacts is observed when transitioning from a fossil-dominated grid (green bars) to renewable electricity (blue bars). For RSB, this transition results in 40–60% reductions across most categories, indicating that cement substitution benefits are preserved only when upstream electricity emissions are constrained. By contrast, in PRSB the levels of reduction are even more pronounced at around 65% for GWP and Acidification due largely to the effects of modification in lowering processing temperature, increased carbon stability, and reduced auxiliary energy demand. This is indicative of the fact that the levels of modification directly impact emission mitigation in the upstream chain. The secondary factor in this analysis is transport distances. As distances of transport are increased from 0 to 1000 km, the results indicate a linear increase in standardized values of about 20–30% over the respective baselines for GWP, Acidification, and Photochemical Oxidant Creation in RSB due to the lower density of RSB diesel consumption per unit of functionality. In addition, PRSB's denser and modified morphology lowers payload-related emissions by ~10–15%, thereby maintaining net environmental savings at moderate transport ranges (200–500 km). Importantly, renewable electricity scenarios consistently buffer transport-induced penalties, keeping PRSB impacts below baseline cement systems across most categories. In addition to the benefits in terms of energy and logistics, PRSB earns system-level credits for plastic waste valorization, mitigating the challenges posed by conventional plastic waste management. Notably, although RSB records tangible baseline gains, its vulnerability to grid load factor and logistics uncertainties is more pronounced. By comparison, PRSB performs more reliably across all scenarios, as indicated by consistently smaller normalized impacts, making it a more scalable low-carbon concrete option in regions where rice and plastic waste are readily available.

Uncertainty analysis

Uncertainty analysis (as shown in Figure S6) using the ReCiPe 2016 Midpoint (H) method was performed to compare plain concrete (A) with biochar-based sustainable concrete (B) at incremental replacement levels (1%, 3%, 5%, and 7%), with results expressed as the probability that $A < B$ or $A > B$ at a 95% confidence interval. The midpoint categories evaluated include global warming potential, fine particulate matter formation, marine eutrophication, human carcinogenic toxicity, and fossil resource scarcity, thereby capturing both climate-related and human–ecosystem health dimensions. Across all replacement levels, the uncertainty distributions exhibit systematic shifts rather than random dispersion, indicating that biochar incorporation introduces directional environmental effects rather than stochastic variability, which strengthens the robustness of the comparative assessment.

For lower replacement rates (1–3%), the uncertainty distribution of the global warming potential indicates a high probability that $A < B$, meaning that ordinary concrete slightly outperforms biochar-modified concrete in some Monte Carlo simulations. This can be attributed to additional processing and transportation costs associated with biochar production, which offset the carbon sequestration effect at lower levels of replacement. However, as the level of replacement increases to 5% and then to 7%, the probability distribution shifts towards $A > B$, indicating a higher possibility of reduced climate impact for biochar concrete. This also implies that after a certain level of replacement, the biogenic carbon sequestration and the mitigation of cement emissions become more significant than the upstream costs. From a mechanistic perspective, biochar can be viewed as a long-term carbon sink due to the persistence of its recalcitrant carbon form, hence its potential in overall greenhouse gas mitigation when applied at adequate levels. The formation of fine particulates has a definite trend for all replacement levels, with a high probability of $A < B$ at lower concentrations of biochar, gradually moving towards equality or slight improvement at higher replacement levels. The initial increase in particulate-related impacts is likely to be related to biomass handling, pyrolysis, and auxiliary energy use in the biochar production process. However, as the concentration of biochar increases, the ranges of uncertainty decrease and move towards lower effects, indicating that the cement dilution effect, in which lower clinker production offsets particulate matter emissions from combustion, becomes more important. The sensitivity of marine eutrophication to the addition of biochar is fairly low, with ranges of uncertainty centered closer to zero for all levels of replacement. For lower levels of replacement, the probability of $A < B$ is slightly more dominant, indicating a slightly greater potential for eutrophication by biochar-treated concrete. This could be attributed to the nutrient emissions during the growth or processing of biomass used as biochar feedstock, especially nitrogen and phosphorus compounds. However, as the replacement levels increase, the distributions approach balance, which shows that the reduction in cement-related upstream emissions offsets the effect of biomass. The human carcinogenic toxicity has a strong and evident pattern for all levels of replacement, with the uncertainty distributions strongly favoring $A > B$. This suggests a high probability that biochar concrete reduces the carcinogenic toxicity impact compared to plain concrete. This is mainly due to the reduced demand for clinker production, which is linked to the release of heavy

metals and carcinogenic substances in the extraction of raw materials. Fossil resource scarcity demonstrates one of the most consistent and statistically robust improvements with increasing biochar replacement. In all cases, the probability distribution clearly favors A > B, with the degree of improvement rising monotonically from 1% to 7% replacement. This is a direct result of substituting energy-intensive, fossil-fuel-based cement production with biochar produced from renewable biomass. As demand for cement falls, so does demand for fossil fuels for calcination and kiln operation, resulting in a corresponding decrease in fossil fuel depletion. Overall, the uncertainty analysis shows that the environmental impact of biochar-based sustainable concrete is dose-dependent: lower replacement rates yield mixed or minimal benefits, whereas higher replacement levels consistently demonstrate superior environmental performance across multiple midpoint categories.

Likewise, the uncertainty analysis of PRSB-based concrete (Figure S8, supplementary material) shows that the progressive incorporation of PRSB reduces environmental impacts compared to conventional concrete, although the extent of reduction depends on the degree of substitution. At the 1% substitution rate, the confidence intervals overlap, which is not surprising, since the clinker component, the major source of CO₂ emissions and mineral depletion, remains relatively unaffected. With 3%, the chances of impact reduction enhanced, especially for global warming and freshwater ecotoxicity. This is because of proportional clinker reduction, which directly reduces calcination emissions and fossil energy demand, while the carbon structure of PRSB adds little to the processing cost. The 5% replacement rate displays the most significant and stable reduction in the impact categories. This rate is likely to achieve the optimal balance between clinker substitution and strength, without the need for any additional binder, and with the maximum possible reduction in emissions. For 7%, there is a greater degree of uncertainty and overlap, which could be attributed to the higher processing energy of PRSB, transport forces, or a slight loss of strength. Therefore, the 5% replacement rate is the most environmentally optimal and technically stable substitution rate in the case of PRSB-based concrete.

Materials and methods

Development of plastic-char composite

The scheme of plastic-char composite development is illustrated in Fig. 12. The plastic-char composite was developed by combining rice straw trash and single-use plastic, specifically low-density polyethylene (LDPE), in a 70:30 (w/w) ratio. The waste materials underwent a thorough washing process using distilled water to eliminate contaminants. Before the blending process, the rice straw and single-use plastic particles were dried at 80 °C to reduce their moisture content. Subsequently, the waste material was chopped into small pieces, preferably smaller than 2 mm, using a shredder. The rice straw and single-use plastic particles were thoroughly blended and subjected to co-pyrolysis in a box-type furnace. The pyrolysis was performed at 800 °C, with a heating rate of 5 °C/min, and was carried out under an inert atmosphere. The nitrogen gas was continuously flushed into the furnace at a rate of 2 L per hour. The pyrolysis process was conducted for 2 h to ensure the formation of a plastic-char composite.

The pristine rice straw biochar was produced using a procedure similar to the one mentioned in the previous section, by pyrolyzing at 800 °C for 2 h at a heating rate of 5 °C/min. Subsequently, both biochar and plastic-char composite were pulverized using a ball mill, sieved to a particle size of less than 150 microns, and stored for further advanced characterization and application for cement replacement in concrete. The pristine biochar and plastic-char composite was designated as RSB and PRSB.

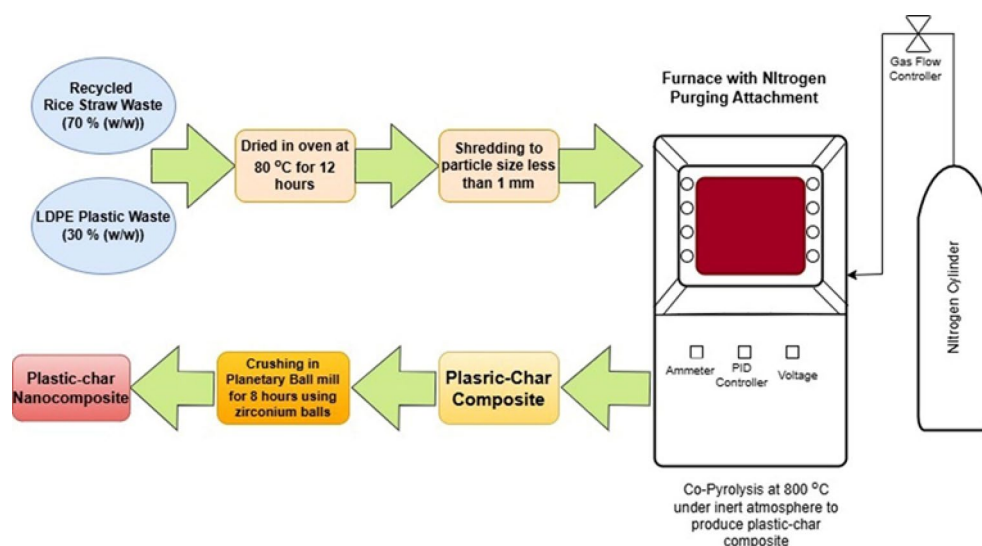


Fig. 12. Schematic flow diagram of plastic-char production using recycled rice straw and single-use plastic waste.

Chemical characterization of plastic-char composite and biochar

The PRSB and RSB were characterized for their physicochemical properties. The elemental analysis was performed using a CHNS/O analyzer (Thermo Scientific, Model Flash 2000, USA), and the proximate analysis was conducted using a thermogravimetric analyzer (NETZSCH, STA 509 Jupiter Classic, USA). The lattice structure analysis was performed using X-ray diffraction (XRD) analysis (Instrument name: XRD, Bruker, Model: D8 Advance, USA) with a Cu K-alpha radiation source (wavelength: 1.5418 Å) at a scan rate of 0.5° per minute. The surface functional groups of the plastic-char composite were analyzed by scanning the surface of the composite in the wavelength range of 4000–500 cm⁻¹ using an ATR-FTIR spectrophotometer (Bruker, ALPHA-II, USA) with an infrared radiation source operating at 12 Volts and 20 Watts. The morphological analysis was performed using a scanning electron microscope (Make: TESCAN, Model: MIRA, Czech Republic) with a resolution range of 1–50 µm and a magnification of 10k–500k×. Furthermore, detailed structural and morphological analysis of both PRSB and RSB were conducted using HRTEM (JEOL, JEM-F200) to determine their surface properties, surface patterns, and the presence of heteroatoms at the nanoscale (2–10 nm). The surface area of both the biochar and plastic-char composite was determined using a BET surface analyzer (Microtrac BELSORP MINI-X). The carbonization level and impurity presence in both biochar and plastic-char composites were analyzed through Raman spectroscopy. (DXR3 Raman Microscope, Thermo Scientific).

Toxicity Characterization Leaching Procedure (TCLP): TCLP was conducted on both RSB and PRSB samples in accordance with EPA Method 1311 to determine the mobility of toxic constituents⁶⁴. All sample extractions were for non-volatile analytes using bottle extraction vessels, which are acceptable for solid matrix materials like the biochars studied herein. Briefly, approximately 100 g aliquots of ground RSB and PRSB (particle size < 1 mm) were weighed to verify that >0.5% solids content would be attained by pressure filtration to 50 psi through 0.22 µm binder-free filter paper. An extraction fluid consisting of 5.7 ml of glacial acetic acid (pH 4.93) and 64.3 ml of 1 N NaOH diluted to 1 L was used in the extraction case. RSB and PRSB samples were placed in 500–600 mL borosilicate glass bottles, with extraction fluid added at 20 times the solid weight. Bottles were rotated end-over-end at 30 ± 2 rpm for 18 ± 2 h at ambient temperature. Post-extraction, slurries were filtered through new acid-washed 0.6–0.8 µm glass fiber filters using positive pressure (up to 50 psi increments), yielding the TCLP extract as the filtered liquid phase. The TCP extracts were immediately acidified to a pH below 2 using 1 N HNO₃. The leachates were then analyzed for heavy metals using ICP-OES (5800 ICP-OES, Agilent).

Mix design and development of plastic-char composite substituted concrete blocks

Ordinary Portland Cement (OPC) of Grade 43 was procured from a commercial supplier and used consistently across all formulations, in compliance with the Indian Standard specifications for structural concrete corresponding to an M30 mix. The control concrete was designed following the M30 mix design protocol, comprising OPC (Grade 43), fine aggregates, coarse aggregates, and potable water, with a target mean compressive strength of 38.25 MPa to account for design margin above the characteristic strength. For the development of biochar- and plastic-char-composite-modified concretes, cement was partially substituted on a weight basis with rice-straw biochar and plastic-rice-straw biochar (PRSB) composite at replacement levels of 1%, 3%, 5%, and 7%. A quantified and repeatable protocol has been specified for dispersion. The determined quantity of PRSB/RSB was pre-dispersed in mixing water using mechanical stirring at 1000 rpm for 5 min and further treated with ultrasonication for 3 min at 300 W to minimize the agglomeration. Then, the obtained suspension was added to the cement–fine aggregate blend during mixing. The adopted protocol resulted in visually uniform distribution without segregation and was followed for all mixtures consistently. All other mix ingredients were kept constant, including aggregate grading and water content, to clearly isolate the impact of cement substitution on hydration, microstructure evolution, and mechanical behavior. The replacement scheme ensured that the amount of char (in terms of mass) was included in all mixes by directly reducing the cement portion and incorporating the equivalent quantity of the char-based material, thereby maintaining a uniform level of total binder in each mix. The specific amounts of cement, biochar (or plastic-char) composite, fine and coarse aggregates, and water used for the control and modified concrete mixes are given in Table 3. This mix design framework enables a systematic assessment of the influence of char-based binder substitution on concrete performance while maintaining comparability with conventional M30-grade concrete systems.

Mechanical strength analysis

The compression strength of the plain, biochar, and plastic-char-supplemented cubes was estimated using the Indian Standard Procedure (IS 516: 1959)⁶⁵. Plain and biochar-based concrete cubes were made with dimensions of 150 × 150 × 150 mm in triplicate. Concrete mixtures were cooled at room temperature first. After 24 h, the concrete cubes were removed from the mold and cured for 3, 7, and 28 days. After curing, the cubes were air-dried to eliminate any water. After cleaning with sandpaper, the cube surface was tested for compressive strength using a digital compression testing machine (CTM) at a rate of 0.2 MPa/s until failure. Flexural strength measures the ability of concrete to resist bending under load, and in this research, the Flexural strength of the beam was measured according to the Indian Standard (BIS 516:1959)⁶⁵. Test specimen beams, 100 mm × 100 mm × 500 mm, for plain, biochar, and plastic-char composite-based concrete were cast in triplicate using the mix design in Table 1. After 24 h, the beams were removed from the mold to achieve minimal handling strength and then cured internally in water for 3, 7, and 28 days. After curing, the beams were placed horizontally in the flexural testing machine with the tension face at the bottom and supported by two rollers. The computerized Universal Testing Machine provided a steady load at 0.7 N/mm²/min at the midpoint for testing. The split tensile strength of regular concrete, biochar, and plastic-char composite-based concrete was evaluated according to the Indian Standard (IS 516:1959) protocols. To comply with IS 516, 100 mm × 300 mm test specimens were produced using the mix design in Table 1 and placed in molds. The test specimens were removed from the mold after 24 h and placed in 27 °C water for 3, 7, and 28 days to cure internally to full strength. The test specimens

were then removed from the water bath and cleaned during curing. Strength was measured by horizontally positioning test specimens in a compression testing equipment. Radially along the cylinder's diameter, 1.2 MPa per minute was applied until the test specimen failed.

Concrete blocks are tested for permeability to see if they can resist water intrusion. This test is essential for concrete fracture resistance and durability. The Indian Standard, IS 3085:1965, water penetration test assessed concrete cube permeability⁶⁶. The permeability test was performed on 150 × 150 × 150 mm concrete cubes using the mix design in Table 1. These cubes were cured in 27 °C water for 28 days. The plain and biochar-doped concrete blocks were carefully cleaned and put in the permeability equipment for experiments. Except for the top, the cubes' sides were sealed to prevent water intrusion. The test specimen received 0.7 MPa (around 6 kg/cm²) pressure from a water tank-connected testing device. Each specimen underwent a 48-hour permeability test. The cubes were then removed from the test assembly and tested for water penetration.

Post-mechanical testing characterization

Following mechanical testing, samples from the plastic-char composite concrete blocks were collected and characterized using advanced techniques to elucidate the mechanisms associated with variations in the mechanical strength of biochar and plastic-char composite concrete materials. XRD analysis was conducted to examine alterations in the lattice structure and mineralogical composition of the modified concrete material. XRD characterization was performed with a Bruker D8 Advance X-ray Diffractometer, utilizing a Cu K-alpha radiation source with a wavelength of 1.5418 Å and a scan rate of 0.5° per minute. Scanning electron microscopy was employed to evaluate alterations or variations in the morphological characteristics of the concrete material produced by substituting biochar and plastic-char composite material. Prior to SEM analysis, the concrete sample was dried and pulverized into a fine powder, followed by a 10 nm gold coating to enhance electron conductivity during the analysis. The analysis of biochar and plastic-char composite-based concrete samples was conducted within a scale range of 50 µm to 500 nm, utilizing a magnification capacity of up to 500 K×.

Response surface methodology and optimization

The optimization of cement substitution in M30 grade concrete using a plastic-char composite involved five steps: (i) identification of dependent and independent variables, (ii) selection of the design model, (iii) data processing and execution of experiments according to the model design, (iv) model fitting with experimental data and statistical validation, and (v) development of surface response curves and the optimization process. Using the Central Composite design, which offers high accuracy and low experimental runs, the study investigated mechanical parameters based on cement replacement and substitute materials (pristine biochar and plastic-char composite). RSM was performed using Origin 2023b (student licensed version). The two-level factorial design of the CCD had 2k factorial points, with the center point indicated as 0 and factorial points from the center as +1 or -1. The number of experimental points is roughly 14 using the CCD equation ($N = 2^k + 2k + n$), where k is the number of design elements, N is the number of experiments, 2^k represents the factorial points, 2k represents the axial points, and n is the number of repetitions⁶⁷. The following second-order polynomial response surface model (equ 1) is used for the RSM and optimization process⁶⁷:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1^2 + \beta_4 X_2^2 + \beta_5 X_1 X_2 + \beta_6 \text{Biochar}$$

Where Y is the response factor (mechanical properties of concrete), β_0 is the overall mean response constant, β_1 , β_2 , β_3 , β_4 , β_5 , and β_6 are the regression model coefficients. Model adequacy was assessed through analysis of variance (ANOVA), evaluation of the coefficient of determination (R^2), adjusted R^2 , and the lack-of-fit test at a significance level of $p < 0.05$. The detailed polynomial equations used for various mechanical strength parameters are shown in Table S1, and the associated ANOVA Table in Table S2 (a–c) (supplementary materials).

Life cycle assessment analysis

The life cycle assessment aims to investigate the environmental performance and consequences of traditional concrete and plastic-char composite-partially replaced sustainable concrete. Figure 13 illustrates how the cradle-to-gate method defines the boundary of the life cycle assessment system. The Life Cycle Assessment (LCA) of sustainable concrete made with a plastic-char composite was conducted using the licensed version of SimaPro 9.6 (Educational software) and the Ecoinvent database. All input parameters were calculated per cubic meter of concrete, with 1 m³ serving as the functional unit. For analytical purposes, raw materials such as OPC cement, fine aggregate, and coarse aggregate were chosen from a database in the Indian setting. The electrical supply was sourced from the southern Indian national grid, and transportation was carried out using a freight truck with a capacity of less than 32 tons under the rest of the world (RoW) scenario, covering a distance of 500 kilometers for raw material shipping. The truck is assumed to be at full capacity during its delivery travel, and its return trip is not included in the model. Additionally, emissions from the system boundary were examined, which included air emissions, water emissions, land contamination, and fly ash formation. The waste disposal scenario was not factored into the LCA study. Finally, a sensitivity analysis was performed, which included all cement replacement percentages utilizing the 'plastic-char composite' at 1%, 3%, 5%, and 7% (w/w). The environmental effect of plastic-char composite-based sustainable concrete was assessed using the ReCipe Mid-point H and IPCC GWP (500-year timeframe) analytical techniques.

Statistical analysis

All data related to the mechanical testing of concrete specimens was collected by triplicating the tests. The collected data were statistically analyzed to calculate the mean and standard deviation. Subsequently, the test

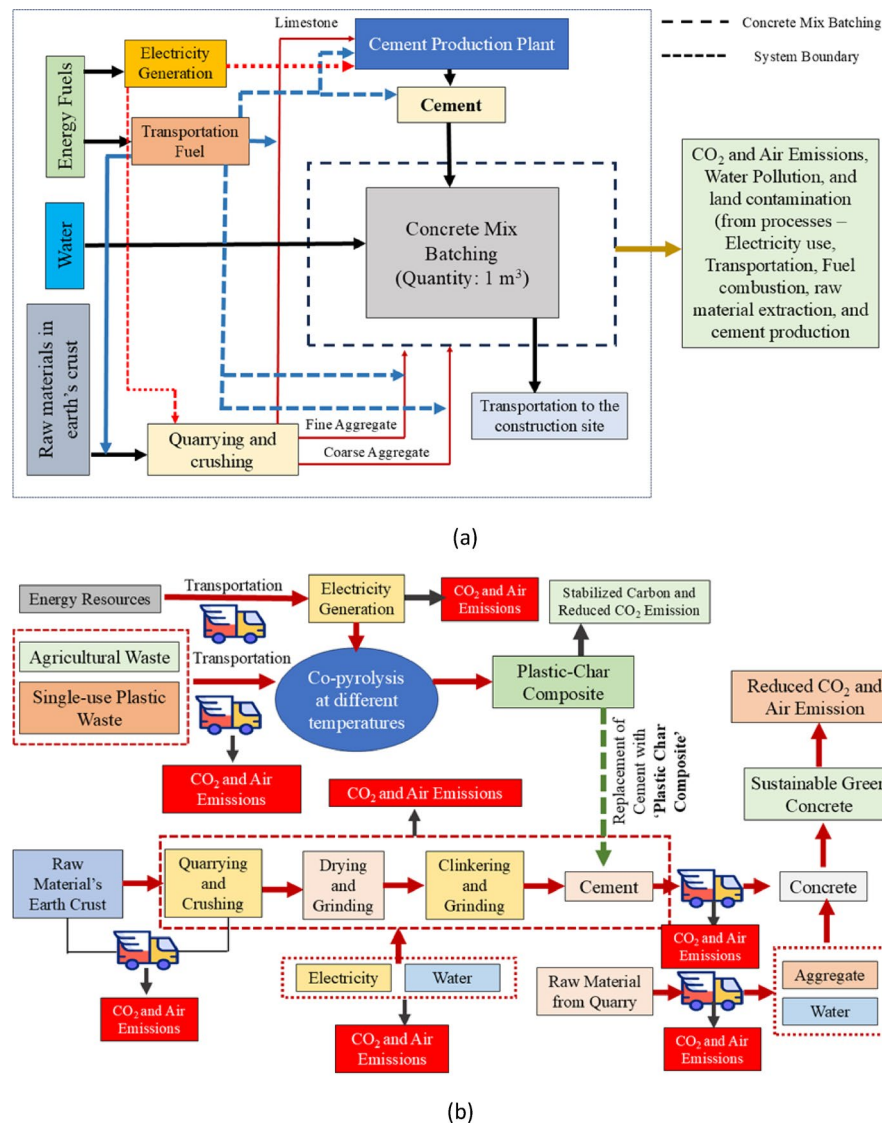


Fig. 13. (a) Life cycle assessment (LCA) of concrete mix preparation using Ordinary Portland Cement (OPC), fine aggregate, coarse aggregate, and water; (b) LCA analysis of plastic-char production using the agricultural waste and single-use plastic waste and development of sustainable and green concrete by partially replacing cement using the plastic-char composite @ 1, 3, 5, and 7% (w/w).

data were analyzed using a one-way ANOVA to assess significant differences in the mean mechanical parameters across the test groups.

Conclusions and future prospects

This work revealed that a plastic char composite can be effectively produced by co-pyrolysis of single-use plastics and rice straw waste, resulting in material characteristics that are significantly distinct from those of virgin rice straw biochar. A thorough characterization utilizing SEM, XRD, FTIR, and EDX demonstrated that the plastic char composite displayed: multilayered structures with enhanced surface area and numerous micropores, in contrast to the macro-pore-dominated structure of pristine rice straw biochar, elevated carbon content and a more graphitic nature as verified by XRD and EDX analyses, and a greater prevalence of functional groups, especially carboxylic and aromatic alkene groups, as indicated by FTIR spectra. When used as a partial substitute for cement at 1–3% (w/w), both pristine biochar and PRSB enhanced the mechanical properties of concrete; however, the plastic char composite consistently outperformed rice straw biochar. Concrete using plastic char composite replacement exhibited superior compressive, flexural, and tensile strengths, as well as diminished permeability and accelerated internal curing. Optimization modelling revealed that the maximum mechanical strength was achieved with a plastic char composite substitution rate of 3.88% (w/w), in close agreement with experimental findings. The Life Cycle Assessment (LCA) further substantiated that employing plastic char composite within the recommended replacement range of 1–3% yields significant environmental advantages and mitigates global warming potential. Nonetheless, substitutions above 5–7% may jeopardize these benefits

due to performance compromises. Plastic char composite serves as a viable, low-carbon, and sustainable supplemental cementitious material, allowing for the concurrent valorization of plastic and agricultural waste in the development of sustainable concrete. This study demonstrates the initial practicality of plastic char composite as a sustainable cement alternative; however, further research is necessary to enhance its commercial viability and ensure consistent performance.

Future studies should focus on long-term durability studies to evaluate performance throughout a prolonged service life under diverse environmental conditions, including freeze-thaw cycles, carbonation, and exposure to sulfate/chloride.

- Assessment of high-temperature performance to ascertain structural integrity and thermal stability for fire safety applications.
- Conduct pilot-scale production and field testing to evaluate scalability, transportation, and integration into current ready-mix concrete facilities.
- The economic feasibility evaluation evaluates feedstock availability, energy needs for pyrolysis, and a cost-benefit analysis relative to conventional cement.
- Optimization of co-pyrolysis parameters (temperature, residence duration, plastic-to-biomass ratio) to customise the composite for certain concrete performance specifications.
- Investigating synergistic combinations with extra cementitious materials (fly ash, slag, silica fume) for multi-functional and high-performance concrete.
- Long-term environmental behavior remains an important research direction, particularly with respect to potential microplastic release. Future work should therefore focus on systematic evaluation of microplastic generation and release from plastic-char-supplemented concrete under prolonged alkaline aging, representative of real cementitious service environments (pH 12–13).
- Comprehensive environmental impact assessment, encompassing not just global warming potential (GWP) but also water footprint, land use effects, and possible microplastic emissions throughout the life cycle.

Data availability

Additional research data will be made available by the corresponding author upon reasonable request.

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References

1. Sirico, A. et al. Mechanical characterization of cement-based materials containing biochar from gasification. *Constr. Build. Mater.* **246**, 118490 (2020).
2. Sharma, P., Moudgil, P. & Mehra, D. *Cement Sector in India*. <https://jmkresearch.com/> (2024).
3. Ho, L. S. & Huynh, T. P. Long-term mechanical properties and durability of high-strength concrete high-volume local fly-ash.pdf. *Results Eng.* **18**, 101113 (2023).
4. Cheng, D. et al. Projecting future carbon emissions from cement production in developing countries. *Nat. Commun.* **14**, (2023).
5. CSIRO. Plastic Waste Issue in India. *Australia's National Science Agency* 1 <https://www.csiro.au/en/news/all/articles/2023/december/circular-economy-roadmap-india> (2023).
6. Nayanathara, T., Pilapitiya, P. G. C. & Ratnayake, A. S. The world of plastic waste: A review. *Clean Mater* **11**, (2024).
7. Ritchie, H. How much of global greenhouse gas emissions come from plastics? *OurWorldInData.org* (2023).
8. Abdurrahman, M. I., Chaki, S. & Saini, G. Stubble burning: Effects on health & environment, regulations and management practices. *Environ. Adv.* **2**, 100011 (2020).
9. Kou, S. C., Poon, C. S. & Chan, D. Influence of Fly Ash as Cement Replacement on the Properties of Recycled Aggregate Concrete. *J. Mater. Civ. Eng.* **19**, (2007).
10. Cheerarat, R. & Jaturapitakkul, C. A study of disposed fly ash from landfill to replace Portland cement. *Waste Manag.* **24**, 701–709 (2004).
11. Elavarasan, S. et al. Experimental study on partial replacement of cement by metakaolin and GGBS. *Mater. Today Proc.* **37**, 3527–3530 (2020).
12. Czop, M. & Zniwska-Piekarczyk, B. Use of Slag from the Combustion of Solid Municipal Waste as A Partial Replacement of Cement in Mortar and Concrete. *Mater. (Basel)*. **13**, 1593 (2020).
13. Wang, C. et al. Utilization of fly ash as building material admixture: Basic properties and heavy metal leaching. *Case Stud. Constr. Mater.* **17**, e01422 (2022).
14. Pushpalal, D., Danzandorj, S., Bayarjavkhan, N., Nishiwaki, T. & Yamamoto, K. Compressive strength development and durability properties of high-calcium fly ash incorporated concrete in extremely cold weather. *Constr. Build. Mater.* **316**, 125801 (2022).
15. Alghazali, H. H., Aljazaeri, Z. R. & Myers, J. J. Cement and Concrete Research Effect of accelerated curing regimes on high volume Fly ash mixtures in precast manufacturing plants. *Cem. Concr. Res.* **131**, 105913 (2020).
16. Pushpalal, D., Danzandorj, S. & Bayarjavkhan, N. Compressive strength development and durability properties of high-calcium fly ash incorporated concrete in extremely cold weather. *Constr. Build. Mater.* **316**, 125801 (2022).
17. AlArab, A., Hamand, B. & Assad, J. J. Strength and Durability of Concrete Containing Ceramic Waste Powder and Blast Furnace Slag. *J. Mater. Civ. Eng.* **34**, (2021).
18. Tan, K. H., Wang, T. Y., Zhou, Z. H. & Qin, Y. H. Biochar as a Partial Cement Replacement Material for Developing Sustainable Concrete: An Overview. *J. Mater. Civ. Eng.* **33**, (2021).
19. Maljaee, H. et al. Effect of cement partial substitution by waste-based biochar in mortars properties. *Constr. Build. Mater.* **301**, 124074 (2021).
20. Chen, X. et al. Sludge biochar as a green additive in cement-based composites: Mechanical properties and hydration kinetics. *Constr. Build. Mater.* **262**, 120723 (2020).
21. Mosaberpanah, M. A., Olabimtan, S. B. & Balkis, A. P. Effect of Biochar and Sewage Sludge Ash as Partial Replacement for Cement in Cementitious Composites: Mechanical, and Durability Properties. *Sustainability* **16**, 1522 (2024).
22. Gupta, S. & Wei, H. Combination of Biochar and Silica Fume as Partial Cement Replacement in Mortar: Performance Evaluation Under Normal and Elevated Temperature. *Waste Biomass Valoriz.* **11**, 2807–2824 (2020).
23. Gupta, S. & Kashani, A. Utilization of biochar from unwashed peanut shell in cementitious building materials – Effect on early age properties and environmental benefits. *Fuel Process. Technol.* **218**, 106841 (2021).

24. Gupta, S., Kashani, A., Mahmood, A. H. & Han, T. Carbon sequestration in cementitious composites using biochar and fly ash – Effect on mechanical and durability properties. *Constr. Build. Mater.* **291**, 123363 (2021).
25. Chandra, S. & Bhattacharya, J. Influence of temperature and duration of pyrolysis on the property heterogeneity of rice straw biochar and optimization of pyrolysis conditions for its application in soils. *J. Clean. Prod.* **215**, 1123–1139 (2019).
26. Samal, B., Raja, K., Kumar, B. & Bhattacharya, J. Char from the co-pyrolysis of Eucalyptus wood and low-density polyethylene for use as high-quality fuel: Influence of process parameters. *Sci. Total Environ.* **794**, 148723 (2021).
27. Samal, B., Raja, K. & Kumar, B. Influence of process parameters on thermal characteristics of char from co-pyrolysis of eucalyptus biomass and polystyrene: Its prospects as a solid fuel. *Energy* **232**, 121050 (2021).
28. Raja, K., Bhattacharya, J., Samal, B. & Chandra, S. Inhibitory and synergistic effects on thermal behaviour and char characteristics during the co-pyrolysis of biomass and single-use plastics. *Energy* **235**, 121369 (2021).
29. Chen, H., Rocha, L. A. O., Zhang, H., Xiong, Y. & Zhang, S. Evaluation of char properties from co-pyrolysis of biomass/plastics: Effect of different types of plastics. *Process. Saf. Environ. Prot.* **193**, 228–238 (2025).
30. Raja, K. et al. Optimized production of single-use plastic-Eucalyptus wood char composite for application in soil. *J. Clean. Prod.* **278**, 123968 (2021).
31. Liew, R. K. et al. Oil palm waste: An abundant and promising feedstock for microwave pyrolysis conversion into good quality biochar with potential multi-applications. *Process. Saf. Environ. Prot.* **115**, 57–69 (2018).
32. Janu, R. et al. Biochar surface functional groups as affected by biomass feedstock, biochar composition and pyrolysis temperature. *Carbon Resour. Convers.* **4**, 36–46 (2021).
33. Han, Q. et al. Biochar derived from agricultural wastes as a means of facilitating the degradation of azo dyes by sulfides. *Catalysts* **11**, 1–16 (2021).
34. Chen, D. et al. The mechanism of cadmium sorption by sulphur-modified wheat straw biochar and its application cadmium-contaminated soil. *Sci. Total Environ.* **714**, 136550 (2020).
35. Tao, Q. et al. Enhanced Cd removal from aqueous solution by biologically modified biochar derived from digestion residue of corn straw silage. *Sci. Total Environ.* **674**, 213–222 (2019).
36. Zhang, C. et al. Impacts of temperature on evolution of char structure during pyrolysis of lignin. *Sci. Total Environ.* **699**, 134381 (2020).
37. Monlau, F., Sambusiti, C., Antoniou, N., Barakat, A. & Zabaniotou, A. A new concept for enhancing energy recovery from agricultural residues by coupling anaerobic digestion and pyrolysis process. *Appl. Energy* **148**, 32–38 (2015).
38. Hongthong, S. et al. Enhanced Biochar Production via Co-Pyrolysis of Biomass Residual with Plastic Waste after Recycling Process. *Int. J. Chem. Eng.* (2024). (2024).
39. Adilina, I. B. et al. Understanding the surface characteristics of biochar and its catalytic activity for the hydrodeoxygenation of guaiacol. *Catalysts* **11**, 1–16 (2021).
40. Schmidt, K., Laumer, J. & O'Leary, S. K. The D- and G-branches associated with the Raman spectrum of thin-film carbon. *Solid State Commun.* **360**, 115041 (2023).
41. Vali, I. P. et al. Bamboo and coconut shell based activated carbon: A Raman spectroscopic study. *Mater. Chem. Phys.* **318**, 129240 (2024).
42. Marzeddu, S. et al. Physical-Chemical Characterization of Different Carbon-Based Sorbents for Environmental Applications. *Mater. (Basel)*. **15**, 1–29 (2022).
43. Barkoula, N. M., Alcock, B., Cabrera, N. O. & Peijs, T. Flame-Retardancy Properties of Intumescent Ammonium Poly(Phosphate) and Mineral Filler Magnesium Hydroxide in Combination with Graphene. *Polym. Polym. Compos.* **16**, 101–113 (2008).
44. Improvement, Q., County, P. & Civilization, E. Effects of biochar on soil properties as well as available and TCLP- extractable Cu contents: a global meta-analysis. *Sci. Rep.* 1–12 (2025).
45. Qin, Y., Pang, X., Tan, K. & Bao, T. Evaluation of pervious concrete performance with pulverized biochar as cement replacement. *Cem. Concr Compos.* **119**, 104022 (2021).
46. Qing, L., Zhang, H. & Zhang, Z. Effect of biochar on compressive strength and fracture performance of concrete. *J. Build. Eng.* **78**, 107587 (2023).
47. Sirico, A. et al. Effects of biochar addition on long-term behavior of concrete. *Theor. Appl. Fract. Mech.* **122**, 103626 (2022).
48. Mahmoud, A. A., El-sayed, A. A., Fathy, I. N., Fawzy, S. & Alturki, M. Evaluation of rice husk biochar influence as a partial cement replacement material on the physical, mechanical, microstructural, and radiation shielding properties of ordinary concrete. 1–28 (2025).
49. Akhtar, A. & Sarmah, A. K. Novel biochar-concrete composites: Manufacturing, characterization and evaluation of the mechanical properties. *Sci. Total Environ.* **616–617**, 408–416 (2018).
50. Li, Z., Xue, W. & Zhou, W. Mechanical Properties of Concrete with Different Carya Cathayensis Peel Biochar Additions. *Sustain* **15**, (2023).
51. Khan, I., Saddiqi, H. A., Ahmad, F. & Ullah, A. Mortar performance enhancement by partially replacement of cement with hybrid biochar: An experimental evaluation and machine learning predictions of compressive strength, water absorption and thermal conductivity. *Mater. Chem. Phys. Sustain. Energy*. **3**, 100024 (2025).
52. Gupta, S., Harn, K. & Sze, P. Effect of biochar on mechanical and permeability properties of concrete exposed to elevated temperature. *Constr. Build. Mater.* **234**, 117338 (2020).
53. Intesarul, M., Khan, R. I., Ashraf, W. & Pendse, H. Production of sustainable, low-permeable and self-sensing cementitious composites using biochar. *Sustain. Mater. Technol.* **28**, e00279 (2021).
54. Gupta, S., Kua, H. W. & Pang, S. D. Effect of biochar on mechanical and permeability properties of concrete exposed to elevated temperature. *Constr. Build. Mater.* **234**, 117338 (2020).
55. Abbas, A. A. & Hapa, S. J. Experimental Investigation of Low-Carbon Concrete Using Biochar as Partial Cement Replacement. *Sustain* **17**, 10744 (2025).
56. Anshul Thakur, R. et al. Enhancement of Concrete Performance and Sustainability through Partial Cement Replacement with Biochar: An Experimental Study. *Iran. J. Sci. Technol. Trans. Civ. Eng.* **49**, 3549–3569 (2024).
57. Qin, Y., Pang, X., Tan, K. & Bao, T. Evaluation of pervious concrete performance with pulverized biochar as cement replacement. *Cem. Concr Compos.* **119**, 104022 (2021).
58. Ying, X., Zhao, X., Ye, M., Wang, C. & Zhan, B. Waste rice straw biochar recycled concrete: Carbon sequestration, durability and microstructure. 512, (2025).
59. Xayachak, T., Haque, N., Lau, D., Parthasarathy, R. & Kumar, B. Assessing the environmental footprint of plastic pyrolysis and gasification: A life cycle inventory study. *Process. Saf. Environ. Prot.* **173**, 592–603 (2023).
60. Zhang, Y. et al. Biochar as construction materials for achieving carbon neutrality. *Biochar* <https://doi.org/10.1007/s42773-022-00182-x> (2022).
61. Ye, P. et al. The state – of – the – art review on biochar as green additives in cementitious composites: performance, applications, machine learning predictions, and environmental and economic implications. *Biochar* 1–30. <https://doi.org/10.1007/s42773-024-00423-1> (2025).
62. Sobol, L., Dyjakon, A. & Soukup, K. Dioxins and furans in biochars, hydrochars and torreficates produced by thermochemical conversion of biomass : a review. *Environ. Chem. Lett.* **21**, 2225–2249 (2023).
63. Murtaza, G. et al. A review of mechanism and adsorption capacities of biochar-based engineered composites for removing aquatic pollutants from contaminated water. *Front Environ. Sci* **10**, (2022).

64. USEPA. SW-846 Test Method 1311: Toxicity Characteristic Leaching Procedure. *USPEA* <https://www.epa.gov/hw-sw846/sw-846-test-method-1311-toxicity-characteristic-leaching-procedure> (2024).
65. BIS 516. Methods of Tests for Strength of Concrete. (2004).
66. 3085, I. S. *Method of Test for Permeability of Cement Mortar and Concrete*. (1965).
67. Sankha Bhattacharya. Central Composite Design for Response Surface Methodology and Its Application in Pharmacy. *Response Surf. Methodol. Eng. Sci.* <https://doi.org/10.5772/intechopen.95835> (2021).

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Author contributions

****P.M.:** Writing original draft and experimentation, ****S.C.:** Design of experiment, writing original draft, editing, and project supervision, ****K.V.R.:** Editing and Supervision, ****I.M.:** Logistic support and editing, and ****N.M.M.:** Data analysis, editing, and validation.

Declarations

Competing interests

The authors declare no competing interests.

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